

The Master Principle of EM Overunity and the Japanese Overunity Engines

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Summary

Conservative fields and single-valued potentials are universally assumed in rotary electromagnetic engine design and analysis, as is the absence of timed automatic *asymmetrical regauging*. Conventionally, a change of EM gauge is *symmetrical* and involves only a change of the EM potentials, while the EM force fields remain unchanged. No excess work of the form $\int \Delta F \cdot dr$ will be accomplished upon a system when it is symmetrically regauged to alter its potential, because $\Delta F = 0$. Excess gradient-free EM potential (i.e., regauging) changes the stored energy in the regauged subsystem, and thus is comparable to a *refueling* process. However, unless an excess force is also produced and allowed to perform work on the system, the excess energy "collected" will not be dissipated to increase the kinetic energy of the rotary engine. Consequently the "free excess fuel will not be burned," and overunity will not result.

Customarily, Maxwell's equations (potential form) are *symmetrically* regauged by electrodynamicists to separate the variables. However, symmetry is obtained only by using two simultaneous *asymmetrical* regaugings, such that the resulting excess force fields offset one another. If the standard net symmetrical regauging is replaced by net *asymmetrical* regauging, then (i) excess potential energy can be freely received from the vacuum and stored in the system, and (ii) the excess force(s) produced can dissipate this excess energy to increase the kinetic energy of the rotary engine, thus permitting overunity operation without violating conservation of energy, the laws of physics, or the laws of nonequilibrium thermodynamics. *Equilibrium* thermodynamics—with its infamous second law—does not apply during the asymmetrical regauging because the engine is an open system freely receiving input of excess energy from an external source (the dynamic vacuum).

A schema for using dq/dt-blocking in a current loop containing a source, as a Poynting S-flow generator, is briefly presented and discussed. Since any source function is only to provide S-flow and emf, it follows that the source function alone does not involve dq/dt or J. The energy collection function in the receiving circuits involves dq/dt and the Slepian vec-

tor $J\phi$. By bridging (dq/dt-blocking while passing S, emf, and $d\phi/dt$) an otherwise conductive section between two dq/dt-isolated closed current loops, then voltage, S-flow, and emf may be freely bridged from the primary sourcing circuit onto the receiving loop containing the load. In short, the primary circuit can fulfill the source function of regauging its receiving circuit containing the load, without exchange of dq/dt or J. However, now no load current passes back through the primary circuit to dissipate its voltage source. In this way, multi-loop electrical circuits can perform asymmetrical self-regauging, thereby achieving overunity efficiency.

In a rotary engine with permanent mag-

net rotor, a ready means of *asymmetrically self-regauging* (ASR) a narrow sector of the stator is to utilize a *multivalued potential* (MVP) in that stator sector, so that the rotor suddenly experiences an instantaneous "jump" in its confronting stator potential and associated force field—as well as a change of sign (potential) and direction (force) if so designed—when it enters the multivalued region. In this fashion the potentialized vacuum itself directly and freely injects excess potential (stored energy) into the system during the multivalued potential jump, *refueling* the system. This asymmetrical potential is created with a gradient *a priori*, so that an associated "free working force" also appears.

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By placing the MVP in the normal "back drag" sector and reversing the sign of ϕ (and the direction of its B), the excess force produced therein will accelerate the rotor positively and increase its kinetic energy. This excess stored kinetic energy can then be dissipated as work in a load during the remainder of the rotary cycle. In principle this is no different than the refueling of a conventional gasoline engine, except that in ASR the vacuum's interaction with matter freely furnishes the "fuel."

Asymmetric self-regauging of a rotary electromagnetic engine thus can provide additional free potential energy from the vacuum and utilize (expend) it to provide additional kinetic energy in the engine, creating a permissible *Maxwell's Demon* of special kind.

A highly nonlinear single-valued magnetic scalar potential with radical magnitude changes in a stator region smaller than the finite domains of the rotor can be utilized as a "pseudo MVP," since a rotor domain will experience this rapid alteration of the magnetostatic scalar potential in a single domain as a nearly instantaneous "jump". Even over several domains, a sharply-changed, single-valued potential with a finite rise time can be used if the resulting primary B -field is radially oriented so that no back-drag results on the rotor. Even for back-drag, the jump time dt can be made sufficiently small so that the overall $\int F(t)dt$ "back impulse" becomes negligible, vanishes, or even reverses. The jumped potential can be appreciably higher than that of the next forward tangential stator region and of like sign. In that case a strong tangential force freely results which *accelerates* the rotor and adds energy to it, bleeding off the energy from the change in potential. Consequently, immediately after the jump the rotor can experience a substantial net overall boost out of the pseudo MVP jump region, and increase its kinetic energy, as formally proven by Johnson's permanent magnet gates in actual laboratory force-time measurements every 0.01 sec, and as utilized in Takahashi's electromagnetic engine.

Explanations of the Johnson boosting stator gate and the Takahashi and Kawai engines are presented from the viewpoint of the potentials and asymmetrical self-regauging. The explanations and the overunity mechanism are straightforward once ASR and the pseudo-MVP jump mechanism are understood.

It appears that the Japanese may be developing an extensive line of these overunity electromagnetic engines for powering electric vehicles, with little or no need for externally recharging the vehicle battery. To maintain scientific and industrial viability, the scientific and industrial communities of other nations must focus attention upon the principles of these engines in their own research, develop-

Table 1: Asymmetry and modern physics.	
* Per T.D. Lee, Particle Physics and Introduction to Field Theory, Harwood Academic Publishers, New York, 1981, p. 378-390.	
SYMMETRY AND MODERN PHYSICS • Missing symmetry of matter • Symmetry only of mass and vacuum • Vacuum excitation • Vacuum structures • Spontaneous symmetry breaking • Interaction between matter and vacuum • Possibility of vacuum engineering	
SOME PROVEN ASYMMETRIES • Positive and negative signs of electric charge • Time reversal • Right/left handedness	
STRONG LOCAL ASYMMETRY • Properties of object may differ appreciably for: - Different observers - Different detecting means - One time to another - One position to another	
STRONG LOCAL ASYMMETRY (CONT'D) • Local spacetime is curved • Lorentz invariance of vacuum violated • May be local sink or source • EM gravitational/inertial effects • Translation between virtual and observable • Electrogravitational solitons • Action at a distance • Transmutation effects may exist • Scalar/pseudoscalar field translation	
WHEN SYMMETRY IS VIOLATED • A "non-observable" turns out to be an observable • An invariance is broken • A conservation law or selection rule is broken	
COMMENTS: • A virtual object becomes an observable object • Local spacetime becomes curved, at least to some potential involving that object	

ment, and production.

Introduction

First some background information. In 1980 I wrote an informal (at the time proprietary) paper pointing out the significance of multi-valued potentials for enabling overunity operations of electromagnetic processes.¹ In a 1989 article² we proposed one method (one of Howard Johnson's methods) for an overunity permanent magnet motor without ancillary electromagnets.³ In 1993 we openly proposed a mechanism for obtaining "free energy" from the vacuum in electrical circuits,⁴ and followed that by filing a patent application and a follow-on continuation on the method and various embodiments.⁵ My associates and I followed with patent applications for room temperature superconductivity, Poynting energy generators and enhancers, and other related overunity embodiments.⁶ We are presently struggling to ready four other patent applications in this rapidly emerging overunity field.

In a recent article⁷ we cautiously pointed out potential Japanese overunity motor work.⁸ With somewhat limited information on engine details and no test results of the reported Japanese overunity engine(s), such caution was appropriate. Therefore we assigned a probability of perhaps only 60% that the work was overunity.

Such caution was justified. In the preliminary electric motor scooter demonstrations held for Western engineers in Europe, no one was permitted to examine the actual motor "under the hood," so to speak. Therefore a question remained, because *independent certification tests*—at least by Western scientists and engineers—had not appeared. Nonetheless I felt that 60% probability existed that the

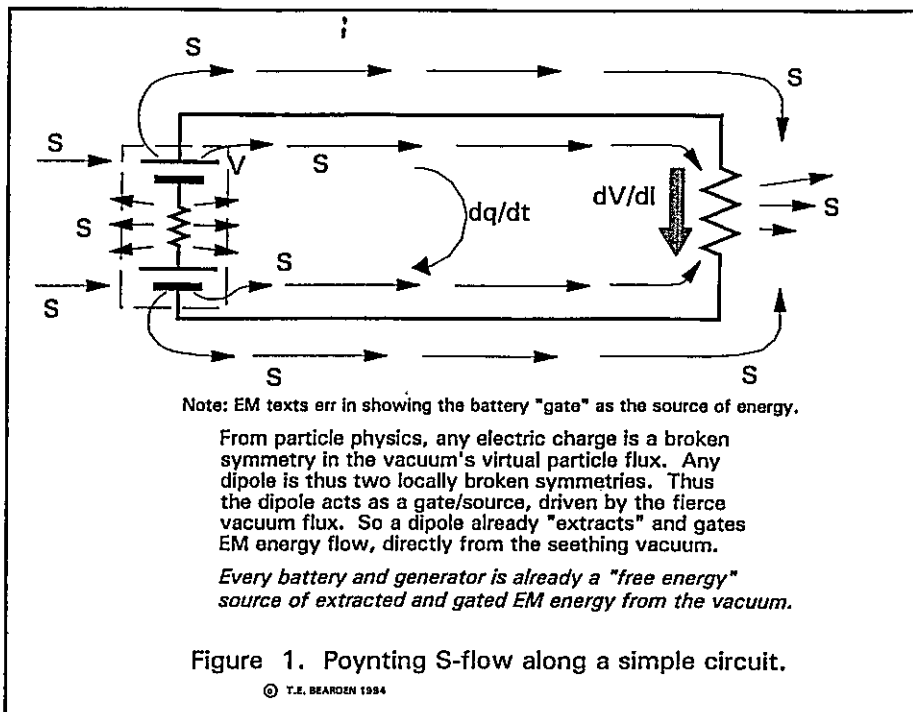
Japanese engine or engines were true overunity engines.

Additional information on Japanese Overunity Engines

Additional information on these Japanese engines is now available. A fundamental U.S. patent⁹ on an overunity engine (test numbers are clearly stated in the patent for overunity coefficient of performance of some 318%) has been issued to Japanese inventor Teruo Kawai. Dr. Eugene Mallove has followed up this startling development by printing an excerpt from the patent, with a commentary,¹⁰ and also the transcript¹¹ of a Fuji Television Network program (some eight minutes) aired on October 20, 1993 over Japanese Television, and dealing with "dream energy" from permanent magnets. Additional tests of the Takahashi motor scooter have been reported, and the Takahashi engine may now be considered as verified overunity.

The Japanese appear to have had a substantially funded, national strategic program in overunity motors and devices for at least twenty years. For example, it is well-known that the Japan is heavily involved in cold fusion research. Also, as early as 1979, a 45 h.p. motor—invented by Takahashi—was being tested in a Mazda automobile.¹² As we shall see, a gauge-theoretic analysis of the Takahashi and Kawai engines and of the Johnson boosting permanent magnet stator gate, supports the technical capability of such engines and devices to produce overunity coefficient of performance, because of their use of *asymmetric self-regauging* (ASR).

The present paper explains some elementary gauge-theoretic concepts needed



to comprehend overunity EM circuits and machines. It also proposes a master overunity principle —asymmetrical self-regauging (ASR)—and explains the application of this principle in overunity electrical circuits, the Johnson permanent magnet boosting stator gate, and the Japanese overunity EM engines.

What is Required to Explain Electromagnetic Overunity

As is well-known in thermodynamics,¹³ any permissible overunity system must involve an open system which receives an input of excess energy from an external source.¹⁴ In other words, the open system must not be in thermodynamic equilibrium. This is important. If the system is open but in thermodynamic equilibrium, it is prohibited from receiving *and storing* any excess input of energy that can then be gated to a load to power it. Therefore the open system in equilibrium cannot exhibit stable overunity coefficient of performance. So the first requirement is to explain how the system breaks thermodynamic equilibrium.

In the case of all three engines, it is important to detail exactly how the excess energy is input into the engine, and show how the system is indeed an open system freely receiving this excess energy from a recognized external source. In short, one must show (i) what the external source is, (ii) how and why the system is indeed an open system, (iii) why it is not in local equilibrium, (iv) precisely how the system receives its "free" input of excess energy from the external source, and (v) what is necessary in order for this excess potential energy to be dissipated as work performed upon the rotary system to increase its kinetic energy.

Energy Can Be Extracted from a Magnetic Dipole

Johnson has now permitted release of the major principle behind his rotor-accelerating magnetic stator—which accomplishes asymmetrical self-regauging by means of a pseudo-multivalued scalar magnetic potential. A patent is in preparation on the process and embodiments. There appears to be no explanation in the Kawai patent or the Takahashi literature as to how the excess energy is arrived at, or what is its exact external source, except for vague reference to "extracting energy from a permanent magnet." To conventional electrical engineers, that kind of statement has long been deemed nonsensical. It is our purpose in this article to show that it is not nonsensical after all, from a gauge-theoretic view.

But first we must break away from exclusive use of conventional EM theory and analysis. *No engine which operates entirely in accord with the standard "closed system" EM textbook model can exhibit an overunity coefficient of performance, because all fields in the conventional theory are already made conservative.* A priori, one must put as much energy *into* that engine to run it, as one can possibly get *back out of it* in the form of work and its losses.

So the first issue is, "How do we create and utilize a nonconservative field in the candidate overunity system?" Some background will be necessary before answering the question.

Asymmetry, Conservation, and the Modern Vacuum

In the standard electrical power engi-

neering model it is assumed that the vacuum has no interaction consequences upon an operating physical system. Consequently many symmetries abound. Indeed, to every conservation law there can be assigned a symmetry. The conservation laws favored by power engineers thus can be regarded from the standpoint of their symmetries. It can come as a profound shock that the assumed symmetry of physical actions in physical systems *does not exist* unless the interaction of the vacuum with the physical matter is included. This is well-known and long since proven in particle physics, but it is absent from electrical power engineering.¹⁵

In material particulate systems, there are three major symmetries—charge, parity, and time—that *individually or even in pairs* need not be obeyed. The requirement is that not all three can be broken simultaneously, in accordance with the well-known CPT theorem of particle physics. In other words, violations of each of the three, and of pairs, have been individually shown; but violations of all three at once have not been found. Violations of each of these symmetries can be most interesting and novel.

For example, local violation of time symmetry yields a local violation of the conservation of energy law, because energy and time are canonical.¹⁶ Electrical charge alone is *not necessarily individually conserved, regardless of what we are ordinarily taught in electrical engineering.*¹⁷ Nature allows charge to become asymmetrical if the vacuum interaction is sufficiently asymmetrical; one must just discover how to do it.¹⁸ Table 1 is a synopsis of some major aspects of asymmetry in modern physics.

Our point is that overunity EM machines must a priori involve one or more broken symmetries. This alone voids any "conventional EM" justification, because it voids any model which ignores the vacuum's interaction with matter and/or which is symmetrically closed. So to ascertain the exact nature of the "external source" that is furnishing the excess energy to an open system with broken symmetry, it is only necessary to realize that *a priori* the source becomes the quantum mechanical vacuum, according to well-established particle physics. Let us examine this a little more closely.

An Electrical Charge Is Already a Broken Symmetry

First, any electrical charge¹⁹ is already a proven asymmetry. According to particle physics and the theory of curved spacetimes, that charge must act either as a gating *source* or a gating *sink* in the fiery flux of the local vacuum. In common terms, that charge is either *extracting* and *gating*²⁰ energy right out of the quantum mechanical vacuum²¹ and thereby acting as a free energy source, or it is gating

energy back into the quantum mechanical vacuum and thereby acting as an energy sink.²²

This immediately puts a different perspective upon an electrical dipole. It is simultaneously similar to a powerful energy spray pump (an energy source) on one end and an energy suction pump (an energy sink) on the other end. Electrical field energy (charge flux) is flowing from the vacuum out of one end of the dipole, and from the other end back to the vacuum—all driven by the vacuum.²³ In general, a Poynting flow must be perpendicular to the field lines, by its definition $\mathbf{S} = \mathbf{E} \times \mathbf{H}$. More conventionally, electrodynamicists assume (for 2-wire circuits) that two S-flows of energy are produced, one from each of the ends of the dipole. There is a net difference between these two flows and therefore a net potential difference between the ends of the dipole.

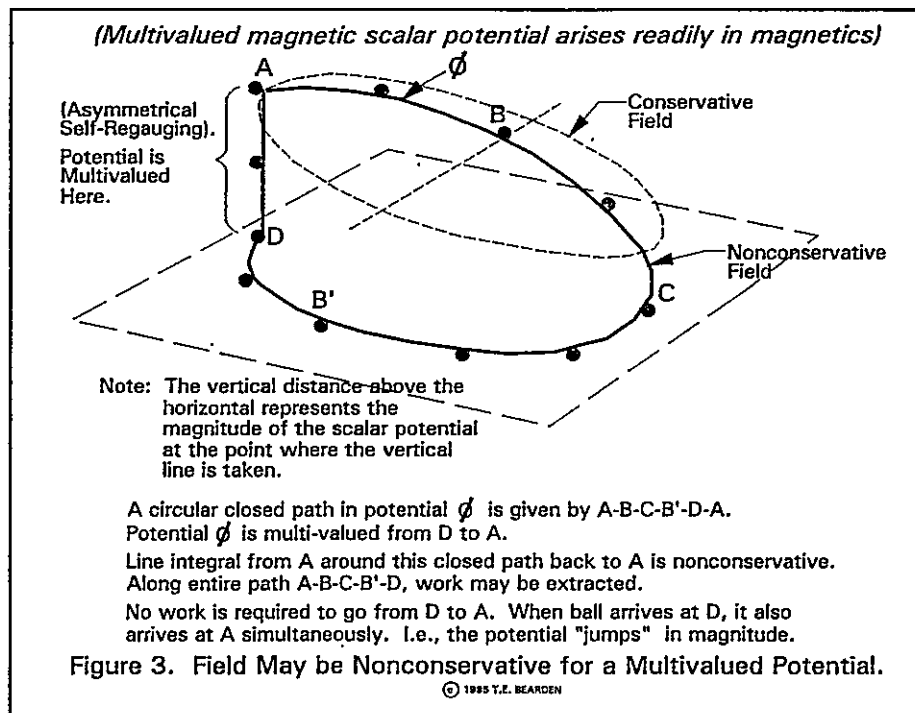
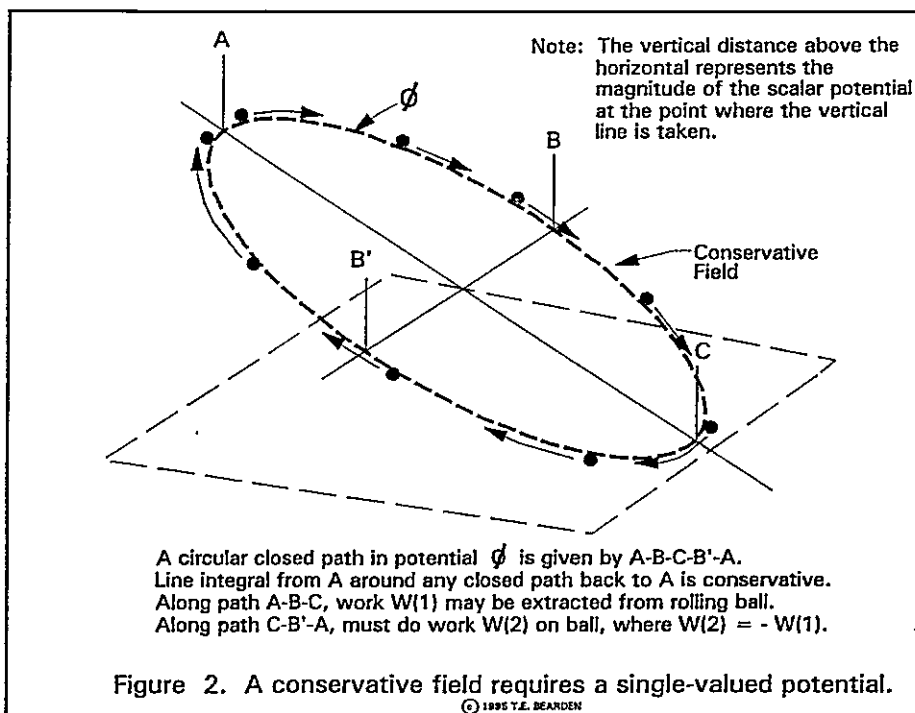
Among leading university EM textbooks, perhaps the best struggle with the flow of Poynting energy in electrical circuits is undertaken by Krauss. Figure 1 shows a partially corrected flow of the Poynting energy in a simple electric circuit, after Krauss and extending his presentation. As can be seen, the flow of Poynting energy originates in the vacuum and returns to the vacuum, contrary to present electrodynamics textbooks. The external circuitry is just the "gating" and "wave guiding" applied to this energy flow, that is freely extracted from the vacuum flux by the dipolar source. Simply eliminate the external circuit, the battery and its internal resistance, but retain the separation of positive and negative charges, and one has arrived at a dipole. *The free flow of Poynting energy from any source is due to this dipole*, not to the overall battery per se.²⁵ The battery or generator is just the mechanism that creates and sustains the dipole.

Every Electric Dipole Is Already a Free Energy Source

Every electric dipole in the universe is already a legitimate, proven, free energy device that directly extracts and gates energy from and to the active vacuum.²⁶ Such a *free energy device* is already included in every battery and generator ever built. It will persist so long as no internal work is done on the charges making up the source dipole, to *destroy* their separation.

One does not have to discover how to extract free energy from the vacuum. One just has to learn how to properly use the free energy extractors *universally* available already.

We now have identified the energetic vacuum as a legitimate primary source of free energy from any dipolar electrical power source. From the seething vacuum, the dipolar source will extract two flows of Poynting energy (we will model it



that way). The external conductors will gate and guide the Poynting energy flows $\mathbf{S}$ — each along with its concomitant emf and voltage — to the external circuit *indefinitely*, so long as the separation of the charges comprising the source's dipolarity is undisturbed.²⁷

A Magnetic Dipole Is Also a Similar Free Energy Source

A similar asymmetry also exists for any magnetic pole (magnetic charge, or magnetostatic scalar potential). A magnetic dipole—any common, ordinary permanent magnet, for example — thus legitimately is

already a proven *free energy-flow generator*. A flow of Poynting energy is moving (by convention) everywhere between its north pole and its south pole, orthogonal to the magnetic field lines. This free flow of Poynting field energy *from the vacuum* will continue as long as that separation of poles exists in the magnet.

One can also represent a magnetic pole as a magnetostatic scalar potential. It is thought that we do not have moving "j Φ " type material magnetic charges in the magnet,²⁸ but we have moving massless $d\Phi/dt$. In the more accurate particle physics view, we have moving virtual photons, moving along the field lines.

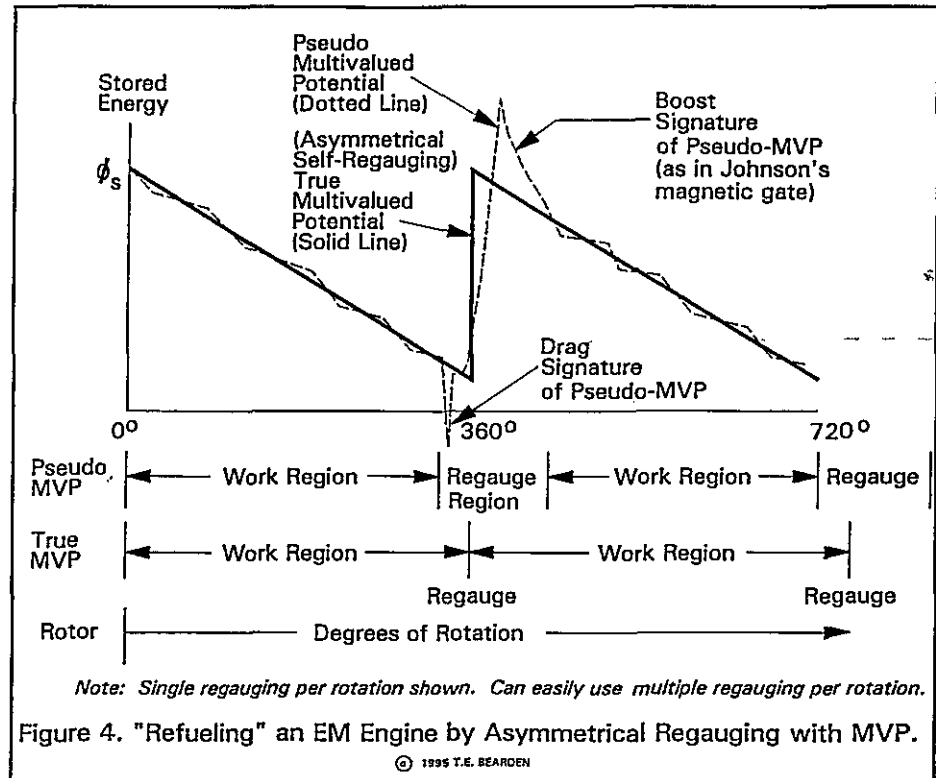
Unknown to present theorists, we also have moving virtual antiphotons, moving along the field lines in the opposite direction. In the dipolar atom, the time-forward photons interact with the time-forward electron shells, while the time-reversed photons (the antiphotons) interact with the time-reversed nuclei.

Regauging a Magnetic Scalar Potential

To asymmetrically "regauge" a magnetostatic scalar potential on a stator so as to enhance the engine's energy and operation, ideally we must create or increase a stator magnetic pole in such a manner that the direct excess magnetic field H from the suddenly injected change in pole strength cannot decelerate the rotor in the regauging region itself. Specifically, the designer must arrange geometry and timing so that a tangential *acceleration* force is created upon the rotor between the regauging region and nextmost regions beyond it. Such a free force which tangentially accelerates the rotor from the regauging region to the nextmost stator region can be highly beneficial. It can "energetically reset" the "spent" portion of the engine, over-ride the otherwise back-drag region and convert it to rotor acceleration, and thereby enable overunity efficiency. In fact, that is the active principle used by both Takahashi and Johnson. [See Figures 10 and 8]

On the other hand, Kawai's engine [Figure 9] creates a tangential force field by a stator electromagnet when it is just forward of the radial flux from a central ring magnet. This produces a conventional accelerating tangential force field, which reduces as the rotor proceeds and the flux becomes aligned with the stator electromagnet. If unchanged, the tangential force component would then reverse in direction and add back-drag to retard the rotor. Just as the tangential force approaches zero and its reversal, the Kawai engine asymmetrically self-regauges by de-energizing the stator electromagnet and resetting that stator coil's magnetic scalar potential back to zero. Hence the asymmetrical self-regauging "quenches" the back-drag field portion. This in turn now makes the overall field situation *nonconservative*. Rigorously, overunity operation is permissible.

Asymmetrical regauging (ASR) is best accomplished by creating the magnetic B -field of the injected pole oriented radially with respect to the rotor pole in the regauging sector, as that rotor pole moves along its tangential path. In that case, no radial work on the rotor system will occur to uselessly drain off the excess energy in the regauged magnetic scalar potential. With the radial B -force from the regauging "bottled up," the injected magnetostatic scalar potential (altered pole strength) from the regauging can readily be made



sufficiently strong as to create a tangential accelerating force between itself and the potential (pole) nextmost in rotation order. Thus the rotor can actually be strongly accelerated "for free" through a region that would otherwise produce back-drag deceleration if asymmetrical self-regauging were not accomplished.

Once the ASR jump of the magnetostatic scalar potential (MSP) is accomplished, the tangential back-drag on the rotor in a permanent magnet motor arrangement can be eliminated or materially reduced — or even reversed so as to aid the rotor's operation — with the expenditure of very little energy in creating the "regauging jump." That is the ASR gauging secret of the Takahashi and Johnson engines. Takahashi employs a breaking of a small current through a radial stator coil in order to induce a momentary, free, very high MSP with its H field radial to the rotor. This produces a large, amplified magnetostatic scalar potential²⁹ "jump" so that the usual "tangential back-drag" force — between the regauging region and the stator region directly ahead — is actually reversed and now strongly aids in accelerating the rotor's movement out of the regauging region. Both the Johnson engine and the Takahashi engine are "convert decelerating drag into accelerating boost" engines, while the Kawai engine is an "eliminate decelerating back drag" engine.

It is important to note that the ASR "jump" region becomes an *energy reset and refueling* region. It is just like refueling a gasoline-powered automobile — by refueling, one resets the stored energy (i.e., the potential) in a subsystem (the gasoline tank) to its desired initial value. The

excess fuel is then expended by the associated excess regauging force, to increase the kinetic energy of the rotary engine, which in turn is dissipated to power the load.

For overunity operation, the engine simply resets (refuels) itself, extracts and dissipates the "excess fuel" energy as work in the load, resets (refuels) itself again, extracts and dissipates more excess energy in the load, and so on.

The Technical Basis of Regauging Maxwell's Equations

For asymmetrical regauging (A-regauging) considerations, we are speaking of altering the Poynting energy flow in a circuit,³⁰ as well as producing excess force(s). This means that the energy is in field energy (E -field and B -field) form, either overtly or "infolded" inside the corresponding scalar potentials,³¹ or both. Consequently, we must analyze Maxwell's equations as we would for radiating energy, rather than employ only the "current and voltage" type circuit analysis conventionally utilized, where the collected energy is sluggishly transported by the Slepian vector $j \Phi$.³² We show in this paper that asymmetrical self-regauging (ASR) allows permissible overunity operation of electromagnetic engines and devices.³³

In Gaussian units, Jackson³⁴ shows that Maxwell's four equations (vacuum form) can first be reduced to a set of two coupled equations in the (A, Φ) representation as follows:

$$\nabla^2 \Phi + \frac{1}{c} \frac{\partial}{\partial t} (\nabla \cdot A) = -4\pi \rho \quad [1]$$

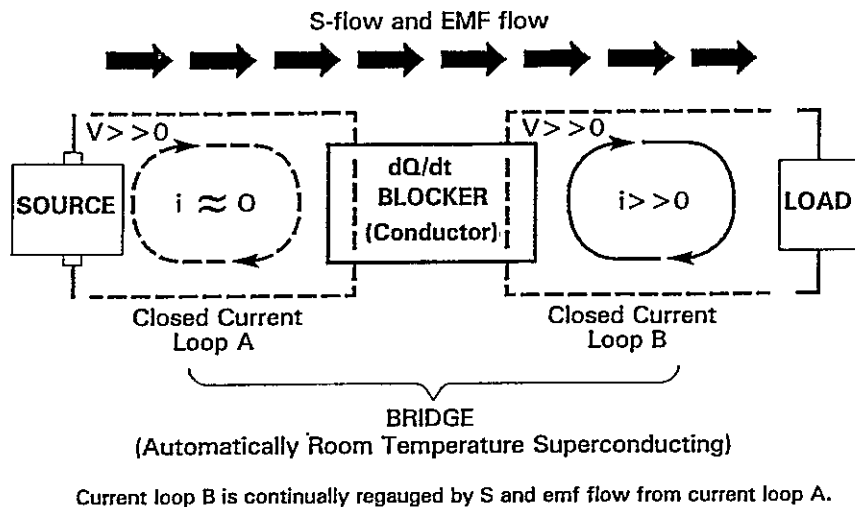


Figure 5. Bridging to enable room temperature superconductivity.

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$$\nabla^2 A - \frac{1}{c^2} \frac{\partial^2 A}{\partial t^2} - \nabla(\nabla \cdot \mathbf{A} + \frac{1}{c} \frac{\partial \Phi}{\partial t}) = -\frac{4\rho}{c} \mathbf{J} \quad [2]$$

The result is two coupled Maxwell equations rather than four. Jackson shows that potentials $\mathbf{A}$ and Φ in these two equations are arbitrary in a specific sense. E.g., the $\mathbf{A}$ vector can be replaced with $\mathbf{A}' = \mathbf{A} + \nabla \Lambda$, where Λ is a scalar function and $\nabla \Lambda$ is its gradient. The $\mathbf{B}$ field is given by $\mathbf{B} = \nabla \times \mathbf{A}$, so that the new $\mathbf{B}'$ field becomes

$$\mathbf{B}' = \nabla \times (\mathbf{A} + \nabla \Lambda) = \nabla \times \mathbf{A} + 0 = \nabla \times \mathbf{A} = \mathbf{B} \quad [3]$$

In other words, the $\mathbf{B}$ field has remained entirely unchanged, even though the magnetic vector potential has been asymmetrically changed. However, if no other change were made, then the electric field $\mathbf{E}$ would have still been changed because of the gradient $\nabla \Lambda$. In that case the net change would be asymmetrical, because one obtained a "free" $\mathbf{E}$ -field which could then do work on the system — either beneficially or detrimentally, depending upon the specific conditions, geometry, and timing. To prevent this excess "free" $\mathbf{E}$ -field from appearing, the electrodynamicists also simultaneously and asymmetrically regauge (transform) the scalar potential Φ so as to offset the $\mathbf{E}$ -field change due to the regauging of equation [3]. In short, they also change Φ to Φ' , where

$$\Phi' = \Phi - \frac{1}{c} \frac{\partial \Lambda}{\partial t} \quad [4]$$

With that additional change, now both the net $\mathbf{E}$ and $\mathbf{B}$ fields remain unchanged,³⁵ even though both potentials

have changed and the fundamental stored energy of the system has changed. Unchanged net force fields just mean that only force-free potentials have been utilized to effect the change in potential energy. It also means that the net summation of the two *individually asymmetrical* regaugings has been *symmetrical*.³⁶

Jackson points out that, conventionally, a set of potentials ($\mathbf{A}$, Φ) is habitually and arbitrarily chosen by the electrodynamicists such that

$$\nabla \cdot \mathbf{A} + \frac{1}{c} \frac{\partial \Phi}{\partial t} = 0 \quad [5]$$

This *net symmetrical* regauging operation successfully separates the variables, so that two inhomogeneous wave equations result that now yield new Maxwell's equations as follows:

$$\nabla^2 \Phi - \frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2} = -4\pi\rho \quad [6]$$

$$\nabla^2 \mathbf{A} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\frac{4\pi}{c} \mathbf{J} \quad [7]$$

Thus the two previously coupled Maxwell equations [1] and [2] (potential form) have been changed to the form given by equations [6] and [7], to leave two much simpler inhomogeneous wave equations, one for Φ and one for $\mathbf{A}$.

Of course this arbitrary *net symmetrical* regauging (net S-regauging) is quite useful for purposes of simplifying the theory and for easing calculations. But its *unquestioning* and rather universal usage has

arbitrarily eliminated the freedom of the system designer to *asymmetrically* regauge the system's potentials, and use the resulting excess free force fields to change the kinetic energy in the system without himself performing extra work upon the system. So we advance the condition for violating this symmetrical regauging process as

$$\nabla \cdot \mathbf{A} + \frac{1}{c} \frac{\partial \Phi}{\partial t} \neq 0 \quad [8]$$

What equation [8] requires is that the two individually asymmetrical regaugings must not be allowed to cancel out each other's freely-produced force fields. Any regauging of the potentials that complies with equation [8] will *a priori* produce one or more excess forces in the system. The *net* regauging will therefore be asymmetrical. By controlling the asymmetrical regauging, the engine designer is then able to control where, how, and when these excess forces appear, and whether or not they are allowed to do work on the system, and whether the work they are allowed to do will enhance the engine's operation or hinder it.

Discussion

Equation [8] has now become the first principle of asymmetrical self-regauging (ASR) for overunity electromagnetic engines and devices. That is, we are now free to A-regauge in a *nonconservative* field manner, where not only do excess potentials appear "for free," but they have gradients so that associated excess force fields also automatically appear for our use in the system. The free force field can drain off the energy in its accompanying free potential's stored energy, and freely perform that much work upon the system.

In applying this asymmetrical regauging approach to overunity, the engine designer is required to pay close attention to the following principles. One should:

(1) Evoke "nonconservative" (asymmetrical) regauging³⁷ in accord with equation [8].

(2) Have the engine assembly and ancillary subsystems perform this noncon-

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Figure 6. Accounting of Poynting-related energy change.

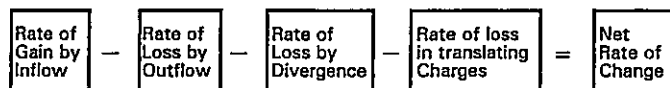


Figure 7. Accounting of Poynting-related rate of change of stored energy.

servative A-regauging in some sector or part of its normal operation.

(3) Insure that any "free regauging force" that is produced will — in doing work upon the system and draining away its associated free potential — increase the kinetic energy of the engine.

(4) Insure that, if the "free" regauging forces that are produced would degrade the system operation, they are bottled up and do nothing, thus avoiding their degradation of the system.

(5) If the regauging forces are bottled up, then the excess potential energy from regauging can be utilized with other nearby non-regauging potentials that differ, to provide additional free engine forces to bleed off the excess regauging potential energy to increase the kinetic energy of the system.

(6) Insure that any additional non-regauging forces so produced will either do helpful work on the system to increase its kinetic energy, or they will also be bottled up so they do no work and therefore cannot hinder the system.

(7) Insure that, in consonance with (1) through (6), at least one free force is formed and utilized to expend the pertinent free potential's energy in doing work upon the engine to increase its kinetic energy.

These above seven considerations we now advance as the primary or master principles of asymmetrical self-regauging of electromagnetic engines and devices for permissible overunity efficiency.

Principles (1) and (2) insure that the system opens itself and receives excess potential energy from the vacuum, and that there exists one or more associated forces powered by the excess potential energy, which the engine designer can use to perform "free input work" upon the

system to enhance its operation. Principles (3) and (4) insure that the free associated regauging forces will either be helpful or they will be nullified. Principles (5) and (6) insure that other forces formed by potential gradients between the regauging potential and nearby potentials will either be helpful or they will be nullified. Principle (7) insures that *at least one* free force and free potential will be utilized to increase the kinetic energy of the engine.

The electrodynamicists have unwittingly *assumed away* the capability of work-free "self-refueling" of an electromagnetic system directly from the local vacuum,³⁸ because they have deliberately designed their systems *not to self-evoke and positively utilize asymmetrical self-regauging of the local vacuum potential*. The Maxwellian EM gauge theory model is thus narrowed to an artificially closed (symmetrical) subsystem of equations that utilize a single fixed gauge, and which exclude from their modeling the free opening of the system and free receipt of excess vacuum potential energy with an associated "acting force" fueled by that excess potential energy. *Maxwell's equations themselves do not require this*, and they do allow self-regauging asymmetrical to "place the system in disequilibrium so that it freely receives and utilizes excess energy from the surrounding vacuum.

By arbitrary habit, electrodynamicists have discarded the major EM overunity mechanism: *Asymmetrical self-regauging without requiring that external work be done on the system*, in order to freely gather in and collect excess potential energy in the system, with associated excess forces, whereupon that excess collected/stored energy can then be used "drive" the forces to freely power the system and its loads. They have also discarded the major overunity corollary of *regauging asymmetrical so that the normal back-drag or back-emf work — usually done upon the system to reduce its kinetic energy — vanishes or reverses sign and becomes bene-*

ficial work done in the opposite sense, to freely increase the energy of the system.

Self-Powering Electromagnetic Engines

All that is required for "self-powering" EM engines directly powered by the vacuum, is that (1) the engine asymmetrical self-regauge to produce $COP > 1.0$, (2) the seven design principles be followed, and (3) positive feedback (from the output) be utilized to its input, in a clamped and controlled fashion, so that part of the engine's output is utilized to run the engine. The feedback can easily be designed and applied, once the fundamental engine is designed for self-regauging and overunity efficiency. The remaining excess output can then be utilized to power loads, while the engine seemingly "powers itself."

Overunity EM system operation via work-free *asymmetrical* regauging has been inherent in the Heaviside/Maxwell equations for a century, but just conveniently and arbitrarily assumed away by limiting the theory to *net symmetrical* regauging.

There is an Infinite Set of Nonlinear Maxwell Equations

Since overunity requires violating the conventional classical EM theory, we must further break our mindset from the typical EM engineering analysis usually applied to motors and generators. Most of the EM theory usually applied by engineers is based upon "the" Maxwell equations. However, when nonlinearities are considered, there is not just *one* set of "the" Maxwell equations available, but in fact there is an *infinite* set of them available.^{39,40} This alone urges great caution when one is tempted to reach sweeping conclusions based upon "the" Maxwell equations and simple nonlinear EM analysis. Modern nonlinear work with electrostatics and Maxwell's theory has shown many things that are not normally taught in university courses, and which are not normally part of the repertoire of the electrical power engineer.⁴¹ For overunity purposes, we are directly interested in the classical EM (CEM) model as a gauge theory, and in asymmetrical rather than symmetrical regauging of the model. In fact, we must focus on asymmetrical self-regauging (ASR) as the way in which an electromagnetic engine can extract excess energy from the vacuum and expend it to power loads.

Three things normally neglected in the conventional EM analysis are (i) any use of a multivalued potential, (ii) any use of non-conservative fields, and (iii) any use of gauge theory and especially *asymmetrical self-regauging* the stored energy in a subsystem at some point in a rotary cycle. We will briefly cover those areas, but only to the extent of what we need to extract from them for our overunity purposes.

Multivalued Potentials and Regauging

An unusual characteristic that can be achieved in nonlinear electromagnetics and circuits is the involvement of a multivalued potential (MVP) and a corresponding "free" asymmetrical change of gauge of the very vacuum/spacetime in which one is working. That is, in one or more regions of the system we can produce automatic "jumps" in the potential energy of the system. If this regauging is also asymmetrical, then excess forces are also produced, which can dissipate the excess free energy to perform work on the rotary system to increase its kinetic energy. This excess kinetic energy then can be easily dissipated in external loads to power them. As stated previously, by convention *gauge transformations*⁴² in electromagnetic theory "consist of certain alterations in the values of those [scalar and vector] potentials that do not result in a change of the electric and magnetic fields."

Modern electrodynamicists utilize the standard potentials A and ϕ , where A is the magnetic vector potential, to represent the electromagnetics in terms of potentials. However, Whittaker has already shown that two scalar potentials can be utilized instead, and still produce all the electromagnetics.⁴⁴ He specifically points out that this is in contradistinction to the standard (A, ϕ) modeling. In short, Whittaker shows that the fields of classical electromagnetics can be replaced by *scalar potential interferometry* of two potentials.⁴⁵

One can symmetrically *regauge* a potential, changing its local value, without necessarily changing the electric and magnetic fields (i.e., without changing the gradients of the potential) in the regauging zone. Or, by asymmetrically regauging, one can change the gradients also. The passage of potential (massless displacement) current $d\phi/dt$ in a circuit, from one isolated loop to another without passage of dq/dt , is a regauging operation, and it is asymmetrical because both an E -field and a B -field are created in the receiving circuit. The Poynting S -flow flows along an equipotential, which is another way of saying that a constant S -flow carries a constant orthogonal E -field and therefore a constant electrostatic scalar potential ϕ right along with it. It also carries a constant orthogonal B -field and therefore a constant magnetostatic scalar potential Φ . If one flows-in excess potential ϕ or Φ , one has flowed-in excess S -flow. And vice-versa, if one flows-in excess S -flow, one has flowed-in excess potential, usually changing the existing potential gradients (i.e., also changing the forcefields).

In the ideal case, no work is required for regauging, whether or not the force fields are unchanged. Work occurs *subsequently*, after any excess force field performs work and dissipated the regauged excess

potential. A multivalued potential asymmetrically regauges the ambient vacuum potential at a multivalued point, establishing free gradients (force fields), and changing the fundamental energy in the system of interest, and thereby permissibly enabling *nonconservative* local fields and *overunity* coefficient of performance. It is very like one of the drawings by Escher, where stair steps descend endlessly because they "close" back via a "regauging" operation upon their starting point, yet are always running downhill.

Regauging by means of scalar potentials only, changes the stored local energy of a system and its local reference potential. That is, at a given "low potential" point or region in a rotary cycle, *a component of the engine can suddenly and freely shift the entire "local vacuum energy" in which that component of the system is embedded*, by means of asymmetrically self-regauging via a multivalued potential. The potential energy of the subsystem freely shifts, and simultaneously provides excess associated forces which can be helpful in using that excess energy to perform work on the engine. In short, the engine *refuels* itself from the vacuum potential.

Pseudo-MVPs and Regauging

In real systems one moves *finite-sized* charges and domains, rather than theoretical "point" charges and point unit north poles, as are conveniently utilized in EM theory. Particularly in magnetics, a multivalued potential can be *emulated* by an ordinary single-valued potential which changes in a very small region with sufficient abruptness.⁴⁶ Such an MVP-emulative use of a single-valued potential to accomplish asymmetrical self-regauging is referred to as a *pseudo-MVP* usage. Either real MVP use or pseudo-MVP use in a system means that one can effectively obtain *nonconservative* magnetic fields and cycles in that system.

A system utilizing an MVP or a pseudo-MVP is inherently capable of overunity coefficient of performance, merely by correctly timing the regauging/refueling operation.

The Conservative Field

In Figure 2 we show by analogy a standard conservative field, viewed from the standpoint of the single-valued scalar potential. We have illustrated a circular track on an inclined board, with a ball rolling around the track. Here gravitation furnishes the potential at point A. The rolling ball passes from A to B to C to B' to A to complete one cycle.

From conventional electromagnetic theory, one of the ways of defining a conservative potential field is well-known to be as follows:

$$0 = \oint \mathbf{F} \cdot d\mathbf{r} = -\oint \nabla \phi \cdot d\mathbf{r} = -\oint d\phi \quad [9]$$

In equation [9] we are referring to the standard line integral of the differential of work about a closed path in a region. Equation [9] specifies that the line integral is independent of the path between any two points P_0 and P_1 , and it holds so long as the potential is continuous and single-valued. For most applications this condition is true, and so equation [9] is a fundamental part of conventional electromagnet-



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Infinite Energy

ic theory, motor design, etc.

Equation [9] may also be visualized as the creation of a potential difference at a point A, with reference to point C, whereby energy flows from the potential to the kinetic system during the travel of the rolling ball from point A to point B to point C. That is, potential energy is changed into kinetic energy of the moving ball, with a consequent "drop" in potential of the ball. Subsequently, however, as the ball continues from C to B' to A, energy flows back from the kinetic energy of the rolling ball to the system potential. If we wished, we could take energy from the system during the travel of the ball along path A-B-C, but we would then have to do equal work back upon the ball during its travel along path A-B'-C. So here we have a *conservative field*. We simply cannot get an overunity coefficient of performance from this field, no matter what path we take. So equation [9] rigorously holds for Figure 2. Note again the requirement that the potential be continuous and single-valued if equation [9] holds and energy is conserved.

Simply put, equation [9] is usually just written as

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0 \quad [10]$$

and, for any system for which equation [10] holds, that system is energy-conservative because the field is energy conservative. Specifically, one cannot get more energy out of such a system than one puts into the system in the first place.⁴⁷

In Figure 2, we can easily see this from the gravitational analogy: Suppose the track and rolling of the ball are drag-free and frictionless. As the ball rolls from A to

B to C, it picks up speed and energy because of its "drop" through a difference in potential between point A and C. In other words, it is rolling "down hill," and the gravitational potential is adding the excess kinetic energy. However, as the ball starts back up hill in its travel from C to B' to A, it will steadily lose energy again, passing that energy back to the gravitational potential. It will just barely reach point A in a totally drag-free system.

For a perfect drag-free system, we could give the ball an initial push from point A, and then it would "orbit" around the track forever. Planets in orbit around the sun are in fact in exactly such a condition, at least to first order.⁴⁸ However, one could not extract any energy from the ball-track system during its fall along path A-B-C, and have the ball continue to roll around the track. So work is added to the ball from the gravitational potential along path A-B-C, and the ball then does work on the gravitational potential along path C-B'-A.

Such a locally conservative field is no use to us in producing an overunity system.

The Non-Conservative Field

But now see Figure 3. Here we show a track that "spirals downward" all the way from A to B to C to B' to D, and where D is directly under the initial starting point A. The gravitational potential adds energy to the rolling ball all along the path from point A to point D. In the ordinary 3-space, the gravitational potential *remains single-valued* along the line from D to A. Therefore we apply equation [9] or equation [10] and the entire field is still conservative. In short, we would have to do as much work on the ball from D to A as we were able to

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gain and extract all the way along A-B-C-B'-D. This is a normal conservative field and it will not yield overunity operation.

However, suppose that we are dealing with electromagnetic scalar potentials instead of gravitational potential. Further, suppose that the potential is *multivalued* from D to A. Now a strange phenomenon occurs. When the system has reached "point D" in its cyclic operation from point A, the potential value itself at D suddenly is altered and "regauged" back to its original value at point A. In short, the value of the potential "jumps" all the way back to the initial value of the potential at point A.

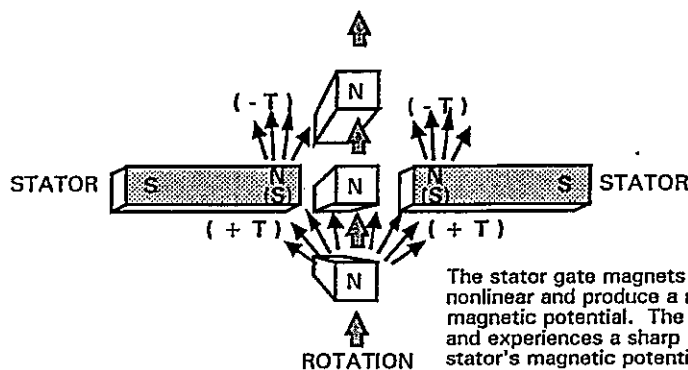
By the gravitational analogy, this would correspond to the ball rolling all the way from point A to point D, and then the vacuum "spacetime" suddenly regauging itself to throw the gravitational potential back up to its initial value at point A. In the physical ball-track analogy, this would also correspond to an external agent suddenly and freely injecting excess energy to *instantly* and *freely* lift the ball from point D to point A.

In that case, so far as we are concerned, we would see the ball just continue rolling always downhill! We could continuously extract energy from it, and we would simply be extracting stored energy from the regauging and refueling by the multivalued potential that had come into being in the region from D to A. We have simply made a *nonconservative field* by use of a multivalued potential to accomplish regauging. Figure 4 shows this "refueling" of an EM engine by MVP ASR.

We accent yet again that no work is required for *perfect* regauging per se, whether the regauging is symmetrical or asymmetrical.

It is possible to make and utilize MVP in electrical machines or in magnetic machines. When we do, we make a non-conservative field in the machine. So we can *permissibly* make an electrical device or a magnetic device that yields overunity electrical performance by ASR. More than a single regauging per rotor rotation can be utilized, depending upon motor design.

To make the so-called "closed-loop"



Effectively a phase conjugation (time reversal) occurs on the exiting edge. The rotor is attracted in; asymmetrical self-regauging occurs. The rotor is accelerated on out. Asymmetrical self-regauging injects free potential energy into the system, from the vacuum. Conservation of energy, the laws of physics, and the laws of non-equilibrium thermodynamics are not violated.

Figure 8. Concept of operation of Johnson's patented magnetic gate.

JPG-5

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free energy machine, all that is necessary is to (i) utilize a nonconservative field cycle by incorporating a regauging and refueling (resetting) sector of the overunity cycle, and (ii) collect a bit of the excess output energy and use it to run the engine. With proper feedback, either an electrical or magnetic overunity machine utilizing a nonconservative field and timed regauging can "run itself" while simultaneously powering a load.

Energy Conservation Laws and Thermodynamics Laws are Not Violated

We strongly point out that use of asymmetrical self-regauging to obtain an overunity system *is* permissible, *does not* violate the conservation of energy law, and does not violate the laws of thermodynamics. It *does* violate the arbitrarily symmetrically regauged classical electrodynamics, as do many other electrical things in physics. It is simply a periodically opened system which receives excess input of energy from the energetic vacuum. In short, its use can provide a "freely self-refueling" engine, analogous to a water-wheel in a flowing stream.

The difference between using an MVP and using a pseudo-MVP is straightforward: For the MVP, one must produce and use an actual multivalued potential. For a pseudo-MVP, one makes a single-valued stator sector potential which, though not actually multivalued, is so nonlinear in that finite but very small region, that the finite rotor system traversing the path experiences the potential in that narrow region essentially as if it *were* a true MVP. In other words, a sufficiently nonlinear single-valued potential can change so sharply in a small region that a finite system perceives the change as instantaneous and thereby perceives the potential as multivalued in that region. The Johnson engine applies the pseudo-MVP principle.

Regauging an Electrical Circuit

In a current loop, whenever one stops the current dq/dt in a conductor but continues to pass voltage as displacement current $d\phi/dt$, one continues to pass the Poynting energy flow given by

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad [11]$$

In short, we pass the flow of potential ϕ (i.e., we pass the voltage) and we therefore change the potential downstream. This *regauges* that part of the circuitry downstream. In one of our patent applications, my colleagues and I described this process as *bridging*. We used bridging to pass the potential from one dq/dt -closed sourcing circuit — where V is large, I is very small, and $[VI \cos \theta]$ is small — to a second dq/dt -closed circuit where V is large, I is large, and $[VI \cos \theta]$ is very large.⁴⁹ In short, we

asymmetrically regauged the potential in the downstream receiving circuit, forming force fields also. That part is work-free flow of excess EM energy into that receiving load circuit. The load is *subsequently* powered by dissipating the regauged potential energy of the load circuit via work performed by the gradients (fields) as they couple to the system and affect it by translocation.

Figure 5 diagrammatically shows our bridging approach that simultaneously produces room temperature superconductivity and is theoretically capable of attaining a system COP >1.0. Work in the regauged load circuit does not "cost" us any degradation of the primary source, and it does not degrade the performance of the source furnishing the potential because it does not require load current to pass back through the source. In short, none of the energy extracted from the primary source is utilized to perform work on the source by driving spent load conduction electrons back through the source against its back emf. Thus no work is done to destroy the separation of charges in the primary source.⁵¹

We now briefly review the background of Poynting theory. Then we show that Poynting S-flow with its corresponding potential flow $d\phi/dt$ (massless displacement current) can yield overunity coefficient of performance in a system wherein a dq/dt -blocked sourcing current loop bridges its S-flow, $d\phi/dt$, and emf to a receiving load loop, where the load loop and the sourcing loop are dq/dt -isolated.

Background of the Conventional Poynting Theory

Poynting and Heaviside developed the energy-flow theory⁵² in EM in parallel, after Maxwell was already dead. Energy-flow does not explicitly appear anywhere in Maxwell's publications; the *notion* that energy flowed through space was exceedingly novel to physicists when developed by Heaviside and Poynting. Circuitry analysis utilizing amperage and voltage were already long established before the notion of energy flow through space was created by the two scientists.^{53,54}

We caution the reader that modern theoreticians are still somewhat confused when attempting to apply the Poynting energy flow to electrical circuitry, and have only done incomplete work in that respect. They have debated the issue for a century. As an example, many engineers believe that the only energy flow in a circuit is by the Slepian⁵⁵ vector $\mathbf{j}\phi$.⁵⁶ They do not realize that $\mathbf{j}\phi$ represents the energy flow density that is collected by the conduction electrons in the wire, from the $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ Poynting flow *outside* the wire, and then sluggishly transported by the potentialized conduction electrons at electron drift velocity. Nominally, $\mathbf{j}\phi$ collects only about 10^{-13} of the S-flow, for subsequent dissipation in the circuitry to power loads and losses.

In short, modern power engineers usually *do not* track the energy flow in an electrical circuit, but only the power and work flow. One very real problem is that power engineers use the term "power" quite differently from physicists, and in two quite contradictory ways. The engineers use "power" to mean the rate of flow of energy that is not dissipating or diverging (i.e., for pure energy transport) and also for the rate of dissipation of the energy. In physics power is rigorously the rate of performing work, which is the rate of *dissipating* or *converting the form* of the energy. Power engineers thus use "power" as an oxymoron. Further, most leading EM texts give Poynting vector application to circuits a very short shift, then leave it without further ado. They do cover radiation of energy into space quite well, however.

Adapting the Poynting Loss Equation for a Special Form of Superconductivity

Let us examine the conventional Poynting energy loss equation. In classical EM energy flow through a unit volume, the Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ represents the amount of EM energy crossing a unit area in unit time, normal to the flow direction of $\mathbf{S}$. A positive value represents EM energy flowing out of the unit volume through unit area, while a negative value represents EM energy entering into the unit volume through unit area. [We will later reverse that tradition]. For a medium that is linear in its electric and magnetic properties, the Poynting theorem⁵⁸ states that

$$\text{div}(\mathbf{S} + \mathbf{S}') + \frac{1}{8\pi} \frac{\partial}{\partial t} (\mathbf{B}^2 + \mathbf{E}^2) + c\mathbf{E} \cdot (i + i') = 0 \quad [12]$$

where $\mathbf{S} = (c/4\pi)(\mathbf{E} \times \mathbf{B})$ is the Poynting vector, and $\mathbf{S}'$ is any vector field whose divergence vanishes; $\text{div}(\mathbf{S})$ is the rate at which the stored field energy is diminishing in the unit volume in question due to a net outward flow of energy;

$$\frac{1}{8\pi} \frac{\partial}{\partial t} (\mathbf{B}^2 + \mathbf{E}^2)$$

is the rate at which the amount of stored field energy in the unit volume is changing, and the expression $c\mathbf{E} \cdot (i + i')$ is the rate at which the electric field does work on all the moving charges, in unit volume, losing energy at that rate. Further, i represents the ordinary gross macroscopic conduction current while i' represents the net

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microscopic current (within the molecules or within the atoms).

In the theoretical case, EM energy which flows along (outside) a wire as S , in an *open* (i.e., dq/dt -blocked) electric circuit, is just such a divergence-free field. If we utilize a multivalued potential, then in the multivalued "jump" region there is just a sudden injection of excess S -flow due to the injection of a different equipotential. Consequently an open circuit — i.e., *one in which the current dq/dt is blocked* — may still pass Poynting EM energy since all the lenergy density low in a circuit exists in the flow of voltage (potential),⁵⁹ and voltage and S may flow without concomitant current. The fact that the EM energy flowing in an electrical circuit does not flow *through* the wire, but *outside* it, is disturbing to many electrodynamicists even today.⁶⁰ The wire serves as a sort of waveguide or railroad track for the energy flow outside. Obviously, if we block the current dq/dt in the wire, we shall block the *expenditure of EM energy as work, from its usual $j\phi$ transporter/carrier*, in that blocked current loop. However, we can still pass the Poynting S -vector flow, and therefore we can still pass EM energy without any dissipation of it in that current loop. Since no work is being performed in the current loop, no work is being done inside the source to destroy it. From this sourcing loop, we can then "bridge" across dq/dt -blocked conductors to a separate receiving loop containing the load.

Bridging provides a special form of room temperature superconductivity where the energy flows by massless energy transport, rather than as $j\phi$. It has an additional advantage of enabling overunity efficiency.

Any Dipole is Already a Source of Free Energy Extracted from the Vacuum

As we previously stated, it is already shown in particle physics that any electric charge is an asymmetry in the violent energy exchange between the vacuum and the charge. Every charge is therefore either a source or a sink. Any dipole is automatically a source on one end and a sink on the other, freely extracting and gating excess energy from the vacuum. This is true for both an electric dipole and a magnetic dipole.

To build a free energy EM source, we do not have to *discover* how to extract and gate excess energy from the vacuum. Every source we have ever built is already doing just that. We only have to discover how to cease using half the extracted energy of a dipolar source, to do work on the internal separation of charges that makes the dipole itself. In other words, we only need to cease "killing" our free energy sources.

dq/dt -Blocking Enables Free Extraction and Use of

Energy from the Vacuum

We now advance some background and a theoretical scheme for asymmetrical regauging of a closed loop load circuit by bridging S , $d\phi/dt$, and emf across a dq/dt -blocked pair of conductors from a dq/dt -blocked sourcing circuit optimized primarily for voltage. No ordinary current dq/dt flows across the bridge in either direction.

The very notion of a dipole already implies dq/dt blocking. In one sense a fixed dipole is a charge-saturated capacitor. That is, even though the two charges are attracting each other, they are not moving together, and there is a potential difference across them. This produces a continuing bidirectional exchange flow of EM energy with every other point in the universe, as Whittaker showed in 1903.

Also, contrary to present electrodynamicists practice, a scalar potential — simple electrostatic voltage — is *not at all* "a static thing" in and of itself. Its *magnitude* at a fixed point is scalar, and the potential is "static" in that it is not changing magnitude! However, internally that "static" potential is comprised of bidirectional EM waves, arranged in precisely coupled pairs, as shown by Whittaker in 1903. If voltage truly were "static," it could never flow along a wire.

A smoothly flowing river may have a static depth magnitude corresponding to any given point on its surface, and the set of those scalar magnitudes is a scalar field. It may also have a static pressure magnitude at any interior point of the river, and that set of scalar magnitudes is also a scalar field. However, no one would suggest that the *river itself* was static. And no one would be astonished that two such rivers could meet, and their internal dynamic structures interfere so that strong wave changes on the surface occurred in the interference zone. No one would state that such "scalar interferometry" was an oxymoron, or that surface waves could not be produced at a distance by those two rivers, which until the interference zone *had* no surface waves.

The scalar field representing the spatial magnitudes of the potential is *not the potential itself*. It is just one *attribute* of the potential. A static magnitude of the potential at each and every point it occupies, does not tell one anything at all as to *what the potential itself actually is, or what its hidden internal structure is doing*. In the wave view, the scalar potential is a harmonic set of hidden bidirectional wave pairs, as shown by Stoney⁶¹ and Whittaker.⁶² In the particle view, the scalar potential is an ordering in the virtual photon flux of the energetic quantum mechanical vacuum.⁶³

Because of its hidden dynamic composition, the electrostatic scalar voltage will "flow" from the charged ends of a dipole if we extend the point charges at the ends to line charges; i.e., if we simply "morph and

stretch out" the end charges. If the resulting *circuit* is open, the voltage will still flow to the new ends, even though (for ideal conductors) no current dq/dt flows. And we already know from Poynting theory that a real flow of EM energy of form $S = E \times H$ continually flows along the conductors.

The primary electrical power source (i.e., the dipole) is *not* dissipated whenever only loss-free field energy density S is extracted from it. Any source of potential is *a priori* a free source of EM field energy density — the open-circuit potential and the S flows *for free* and at no dissipation to the source. The energy is extracted from the vacuum and flows along the blocked (lopen) circuit wires as lossless flow of potential onto collectors (such as electron capacitors). Applying this potential to trapped charges in a "saturated-charge" capacitive collector allows energy to be freely extracted from the vacuum by the source, and furnished to the collector as excess field energy on the blocked electrical charges of the collector.

So here we have extracted another principle: When the dq/dt current is blocked in a conductor, the conductor will still pass S -flow and the flow of potential.

With dq/dt blocked, if voltage (potential) is varied across a coil, normal magnetic field B is created in the coil by the changing E -field. Energy is thus stored in the coil in the form of ordinary B -field.⁶⁴ It does not require moving charges dq/dt to create a magnetic field. *As is well-known, in the vacuum the magnetic field is continually created by the varying E -field, and there are no observable charges q present nor is there any dq/dt present.*^{65,66}

Further, if the storage coil is the primary of a transformer, a correspondingly induced alternating B -field is created in the secondary in normal fashion as the B -field in the primary is produced in AC fashion. In short, *field energy density* flows from the vacuum through the charge-blocked primary and across to the secondary without work.⁶⁷ In the external circuit on the secondary side, charges need not be blocked. In that case the excess field energy density and emf cumulated in the secondary *couples* to the conduction electrons in the secondary circuit. In the secondary closed current loop, the coupling-in (shuttling in) of excess voltage and emf activates the circuit, and drives electrons around the secondary current loop, passing through the load and powering it in normal fashion.⁶⁸ None of this load discharge current passes back through the original source; hence no dissipation of the source occurs, even though the load is powered. So the charge-blocked and energy-shuttling system in this case has become a continuous *free power* system, powering a load without exhausting the primary power source.

In this way, in theory the source will power a load indefinitely. Energy is conserved, but not work. There is no law of physics that requires work to be con-

served! The laws of physics and thermodynamics are not violated. This is an *open* system, receiving and extracting energy from a recognized source (the virtual photon flux of the quantum mechanical vacuum).⁶⁹ The continual bidirectional virtual energy exchange between the vacuum and charged particles in the dipolar separation of charges in the electrical power source provides a continual inflow and outflow of vacuum energy.⁷⁰

By dq/dt-blocking in the external circuit, this exchange energy flux (potential) is gated by the circuit wiring (waveguides) and moved down the circuit without any work being done in the source antenna to dissipate its separation of charges. In the external circuit the energy density is collected as a finite amount of excess energy in a finite collector such as a coil, or a capacitor whose dielectric cannot undergo additional strain.⁷¹ By switching the capacitive collector or by inductive transformer coupling, the collected free energy is shuttled to a separate circuit and regauges that receiving circuit. The excess energy now stored in the regauged receiving circuit is then discharged normally in the load. In either case the energy collection circuit and the load discharge circuit are in separate, isolated current loops. The result is a legitimate open system capable of overunity coefficient of performance (COP), comparable to a heat pump.

Adaptation of the Poynting Theory for dq/dt-Blocked Conducting Circuits

We now examine the Poynting flow when operating a circuit in a charge-blocking fashion. We start by examining the standard Poynting theory for loss of field energy stored in a unit volume, as repeated in equation [13] below.

$$\text{div}(\mathbf{S} + \mathbf{S}') + \frac{1}{8\pi} \frac{\partial}{\partial t} (\mathbf{B}^2 + \mathbf{E}^2) + c\mathbf{E} \cdot (\mathbf{i} + \mathbf{i}') = 0 \quad [13]$$

The Poynting theory of equation [13] addresses the loss of EM field energy from a unit volume. It states that the field energy in a unit volume of interest can be lost by three methods, as previously explained. The three loss terms in equation [13] do not overtly allow a flow of energy into the unit volume, except by the divergence loss term becoming negative and hence a convergence, the field energy loss term becoming negative and hence an inflow, and the work term for translation of charged particles becoming negative, in which case the particles are giving up field energy to the volume.

So as far as electrical circuits are concerned, the standard Poynting equation is an awkward expression, primarily adapted to deal only with energy flow out of unit volume and through unit area thereof.

However, this shortcoming can be remedied by rewriting the terms so that all expressions deal both with field energy loss and field energy gain. For clarity, we use the following word equation shown in Figure 6 for a change in the field energy stored in a unit volume:

Figure 6 shows the word equation used to set up the necessary mathematical relationships. One simply starts with an initial amount of field energy in the unit volume, then adds all gains and subtracts all losses, to arrive at the final field energy stored in the unit volume. By definition, we take the convention that the algebraic sign of gains is positive, and the sign of losses is negative.

Figure 7 shows the word equation used to set up the necessary rates of gains or losses of field energy in a unit volume. By integrating each term over a definite time t , that term becomes an amount of change after that time t has elapsed. By then adding the initial EM energy, the final remaining amount of energy in the unit volume can be ascertained as was shown in Figure 6.

We take the initial field energy stored in the unit volume of interest as

$$\frac{1}{8\pi} (\mathbf{B}_0^2 + \mathbf{E}_0^2) \quad [14]$$

where the subscript zero means *initial*. We take the amount remaining after the change as

$$\frac{1}{8\pi} (\mathbf{B}_f^2 + \mathbf{E}_f^2) \quad [15]$$

where the subscript f means *final*. The amount gained by inflow is the time integral of the instantaneous rate of inflow, or

$$-t \frac{1}{8\pi} (\mathbf{B}_{in}^2 + \mathbf{E}_{in}^2) \quad [16]$$

The amount lost by outflow is the time integral of the instantaneous rate of outflow, or

$$-t \frac{1}{8\pi} (\mathbf{B}_{out}^2 + \mathbf{E}_{out}^2) \quad [17]$$

We shall assume there is no field divergence loss, since the circuitry wires act as waveguides and *energy transport flow* is confined to the waveguides. The loss in performing work to translate conduction charges is given by the time integral of the instantaneous rate of loss for conduction charge translation, which is

$$-tc\mathbf{E} \cdot (\mathbf{i} + \mathbf{i}') \quad [18]$$

Note that, if we do not consider the internal atomic or molecular charge movement, this reduces to

$$\int_t^? c\mathbf{E} \cdot \mathbf{i}$$

Further, if we block the charge flow $\mathbf{i}$, the remainder of the term reduces to zero.

So with charge-blocking, the instantaneous rate of change of the field energy

$$\frac{1}{8\pi} \frac{\partial}{\partial t} (\mathbf{B}^2 + \mathbf{E}^2) = \frac{1}{8\pi} \frac{\partial}{\partial t} (\mathbf{B}_{in}^2 + \mathbf{E}_{in}^2) - \frac{1}{8\pi} \frac{\partial}{\partial t} (\mathbf{B}_{out}^2 + \mathbf{E}_{out}^2) \quad [19]$$

stored in the unit volume is just. In other words, the situation has reduced to pure energy transport without loss.⁷² Also, among other things this immediately proves that use of dq/dt-blocking accomplishes regauging of the downstream receiving circuit.

We note that the two terms on the right side of equation [19] need not be equal. If the rightmost term is largest, then the stored energy is discharging. If the leftmost term on the right side of equation [19] is the larger, then the stored energy is increasing.

When we regauge the system by injection of excess potential energy, then the leftmost term on the right side of equation [10] is much larger, and the stored energy of the system is rapidly increased. Integrated over the very small but finite "jump" time dt , equation [19] then gives the total regauging energy freely added to the system.

We further point out that equation [19] applies to either an electrical device or a magnetic device, or a combination. If the current-free field energy is being stored in an inductive collector, the $\mathbf{B}$ -field increase is significant. If the current-free field energy is being stored in a capacitive collector, the $\mathbf{E}$ -field increase is significant.

dq/dt-Blocking as Asymmetrical Self-Regauging

With dq/dt-blocking, one may conduct work-free energy flow along electrical circuits without losses. This energy flow is in the form of $d\phi/dt$, or simply put, dV/dt . It automatically involves $\mathbf{S}$ -flow and regauging of the receiving circuitry. It is directly comparable to the flow of open-circuit voltage down an ideal transmission line with an open circuit. From the standpoint of gauge field theory, it also involves a change of gauge in the receiving parts of the system, as we stated previously.

What is actually being conducted is the change in the *rate of virtual photon exchange* between the vacuum and charged particles. If the charges q do not translate (i.e., if their longitudinal movements are blocked), then $\nabla\phi$ (with respect to ground reference) is built up on each of them. That is, each of them is potentialized to the new potential. In this manner the energy coming from the vacuum exchange may be captured and stored on trapped/blocked charges q as excess $\mathbf{E}$.

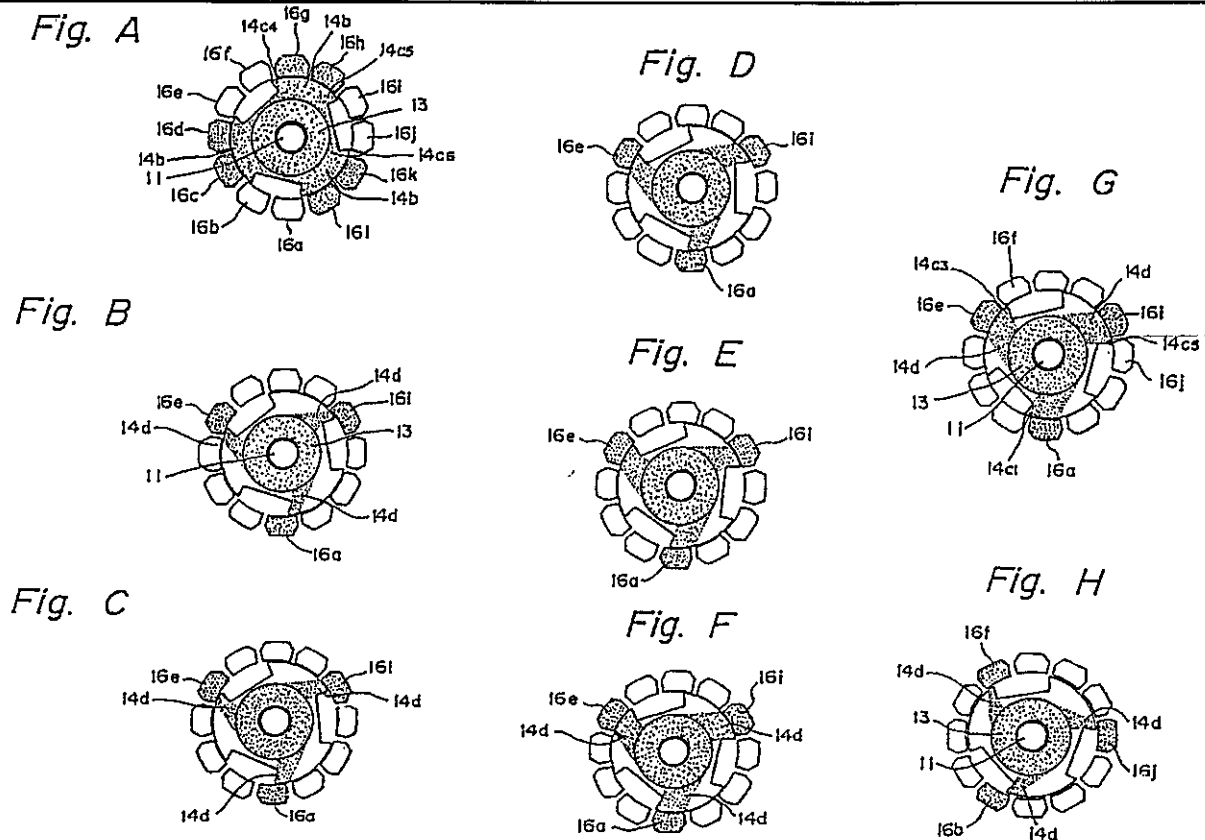


Figure 9. Rotor advance and asymmetrical self-regauging in a typical Kawai engine.

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fields between the potentialized charges and the ground line, and hence as excess field energy. This is a capacitive-type storage. Taking *only* voltage from the power source does not diminish its ability to continue to furnish such voltage. Adding this voltage V to trapped charges q produces an energy storage on those charges of Vq . The collector with its stored energy may then be switched out of the collection circuit and across a load, where it is discharged normally through the load in a separate circuit. The collector can then be switched back to the blocking/collection circuit to collect additional free energy. Obviously this modality allows overunity COP.

The Johnson Force-Producing Magnetic Gate

Johnson's patented force-producing magnetic stator gate utilizes a pseudo-MVP created by a complex nonlinear assembly of permanent magnets. When the rotor polepiece enters the stator gate in attractive mode, it undergoes an asymmetrical self-regauging operation upon encountering the MVP in the middle of the gate.

Figure 8 diagrammatically illustrates the operation of the force-producing gate in Johnson's permanent magnet motor. As

Johnson has shown, by using a pseudo-MVP in his stator gates, an approaching rotor magnet is attracted into a highly nonlinear stator gate region where the MVP is located. When it enters the MVP in the attractive mode, the rotor encounters a sudden dramatic jump in the stator's magnetic scalar potential with a change of polarity. Reversal of the polarity is a *time-reversal* operation, or a phase conjugation operation. In the diagram we have shown the time-reversed aspects.

In turn, this sudden change of magnetic polarity and increase in its magnitude, constitutes an asymmetric self-regauging. This produces both an excess magnetic scalar potential (pole strength) and an associated, sudden, *accelerating* tangential force in the region which would otherwise have been the back-drag region. This accelerating force propels and accelerates the rotor magnet on through the gate and out of it, thus increasing the kinetic energy of the engine. That excess energy can then be taken out as shaft power to power the load.

Rigorous force meter measurements taken at 0.01 second intervals prove that this asymmetric self-regauging by a pseudo-MVP occurs as the rotor passes through Johnson's gate. A representative plot of such force meter measurements is shown as the dotted line in Figure 4.

Johnson thus uses a highly nonlinear magnet assembly of special design to create an MVP in his gate. The MVP produces a "magnetic potential jump" and a reversal of the (otherwise) exiting back-drag on the rotor. In short, Johnson uses an MVP to cause the system to be automatically "refueled" by asymmetric self-regauging, so that it can continue to rotate and power a load. The Johnson engine is thus a self-powering permanent magnet engine utilizing the principles described in the present paper.

The Kawai Engine

Figure 9 shows eight snapshots of the rotor advance of a typical Kawai engine, taken from Kawai's patent.⁷³ This is one end rotor/stator side of a two rotor device, where a similar rotor/stator device is on the other end of the central shaft 11. In Figure 9A, polepiece 14 has three outward teeth 14b dispersed equally around the circumference, alternated with three notches. An end magnet 13 provides the source of flux passing through the polepiece. With the electromagnets de-energized, their core materials 16c, 16d, 16g, 16h, and 16k, 16l are shown shaded, by flux from central magnet 13 outwards through teeth 14b.

In Figure 9B, electromagnets 16a, 16e, and 16d are energized. The shaded area

shows the sharp convergence of the flux from magnet 13 through polepiece 14 and the edge of teeth 14b. Since the electromagnets are magnetized in attracting mode, the rotor will experience a torque tending to widen the flux path from magnet 13 to the activated electromagnets. Thus a clockwise torque exists on the rotor, and it will start to rotate clockwise.⁷⁴ Note also that each electromagnet is operating independently of the other two.

As shown in Figures 9C, 9D, 9E, and 9F the rotation of the rotor continues clockwise, widening the connecting flux path to the three activated electromagnets. During this time the torque on the rotor is clockwise.

In Figure 9G, the flux path to the activated electromagnets is fully widened. Also, the leading edges of the three teeth are just beginning to enter the domains of the next electromagnets 16j, 16b, and 16f.. This is getting similar to the original position shown in Figure 9B. Consequently, the electromagnets 16i, 16a, and 16e are deactivated, and electromagnets 16j, 16b, and 16f are activated. Asymmetrically, this self-regauges and resets the engine back to the original starting position in Figure 9B. The action cycle begins anew. As can be seen, in each complete rotation of the shaft, each of the three teeth of the rotor will be regauged 12 times. So 36 total regaugings/resetting/refuelings are utilized per shaft rotation.

In each stator coil, at energization a tooth is just entering that coil. Energized in attractive mode with respect to the ring magnet around the shaft, the flux in the polepiece "jumps" from fully widened flux (and small or vanishing radial torque on the rotor) to angled and narrowed flux (with full radial clockwise torque on the rotor). As previously explained, the narrowed flux and its angle exert a clockwise accelerating tangential component of force upon the rotor. Each coil is de-energized prior to beginning to exert radial back emf (which it would do if it remained energized as the trailing edge crossed it and again narrowed the flux path). So the Kawai engine uses normal magnetic attraction to accelerate the rotor for a small distance, then asymmetrically self-regauges to zero attraction to eliminate the back-drag portion of the attractive field. It regauges to zero the back-drag as the "RESET" condition.

For appreciable power and smoothness, the Kawai engine uses an extensive number of asymmetrical self-regaugings per axle rotation, being 36 times on each end, or a total of 72 for the two ends. The forcefield of each coil, accompanying its increased magnetostatic scalar potential, is oriented radially inward, so that radial work cannot be done by the coil on the rotor because the rotor does not translate radially. Advantage is taken of the initial clockwise acceleration force initially pro-

duced, and ASR eliminates the counter-clockwise drag or "decelerating" force that would be produced without it.

The major benefits of the Kawai arrangement are that (i) a large number of ASRs occurs for a single rotation of the rotor assembly, enabling high power-to-weight ratio, (ii) each electromagnet is energized only when positively contributing to the clockwise torque that drives the rotor, and (iii) each coil is de-energized to RESET the system the drag-free condition during those periods when the coil would otherwise create back-drag (counterclockwise torque) if it remained energized.

So the Kawai engine delivers what it advertises: It dramatically reduces or eliminates the "back drag" from the fields of the stator electromagnets, because there are no fields activated in the electromagnets during the back-drag sectors of rotation. A conservative field cycle is one in which the back-drag is equal to the forward boost. Eliminating the back-drag portion of the cycle is a form of asymmetrical self-regauging. Note that again it was accomplished by a change in the magnetostatic scalar potential, where the potential is reset to zero by the de-energized coil during the back-drag portion of an otherwise conservative cycle. The Kawai engine therefore uses asymmetrical self-regauging and nonconservative fields in order to permissibly achieve overunity operation.

Because of the numerous ASR regaugings and back-drag elimination, this engine definitely can provide a COP>1.0. Placed in an electric vehicle with necessary switching circuitry and ancillary equipment,

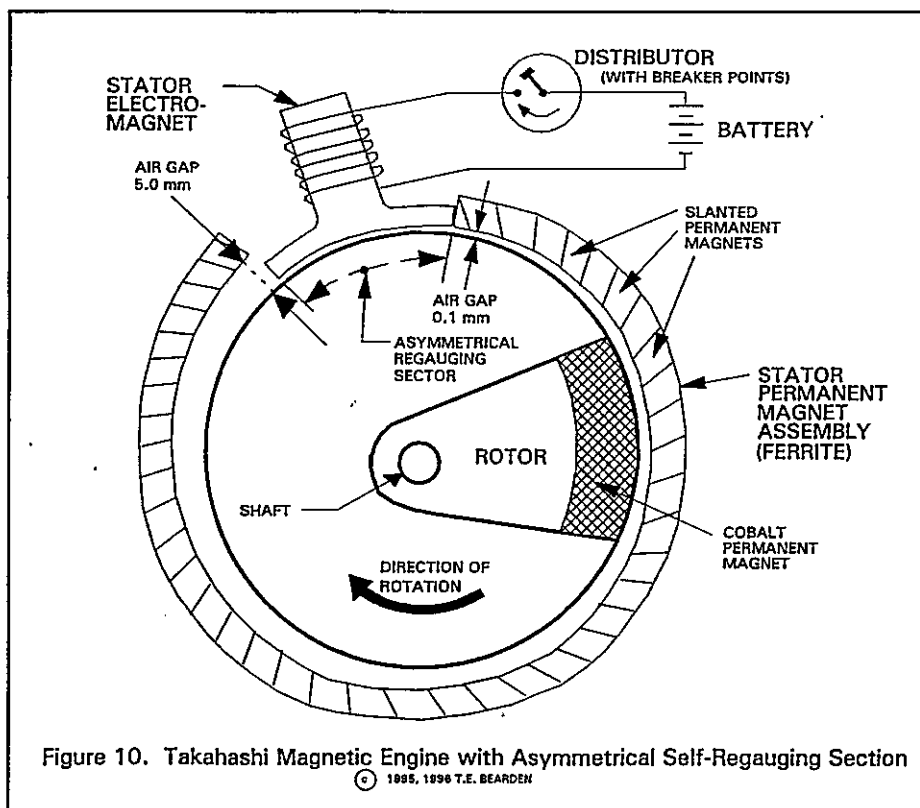
a properly designed Kawai engine and its derivatives should be capable of starting from a single ordinary battery, then powering the vehicle agilely, powering the accessories, and recharging its own battery — all three simultaneously. And in so doing, it complies with all the laws of physics and thermodynamics.

The Takahashi Engine

See Figure 10 for the principle of the Takahashi engine. In the engine a set of permanent magnets, each at an angle to the various radial lines of the device, comprises a slightly widening spiral stator that is "almost" circular but not quite. A circular rotor is mounted inside this spiral stator. An end gap exists in the stator as shown, so that the stator is not a complete closed ring. The direction of rotation for the rotor is clockwise as shown. For demonstration of the principle, the beginning air gap is 0.1 mm and the ending air gap is 5 mm.

A permanent magnet is mounted along the perimeter of an angular sector of the rotor. It is magnetized, say, with the north pole facing radially outwards, and the south pole facing radially inside. In the stator, the permanent magnet north poles are facing radially in toward the rotor, but at an angle, and the south poles are facing radially outside but at an angle.

Thus *tangentially* the north pole of the rotor is in a nonlinear magnetic field, and it will experience a clockwise force and acceleration from position 1 (where the air gap is the minimum) to position 2 (where the air gap reaches maximum).



If this were all there was to it, the Takahashi engine would not be overunity because the tangential field is conservative. When the rotor crossed the end gap in the stator between point 2 and point 1, very sharp and dynamic braking work would be done back upon the rotor magnet by the field of the stator magnets at point 1. This braking work would precisely equal the amount of dynamic acceleration work that was done in accelerating the rotor magnet from position 1 to position 2. For an absolutely frictionless machine with no losses, the coefficient of performance (COP) would be 1.0. Since any real machine will have at least some friction and drag, the actual COP would be less than 1.0.

Let us now utilize the notion of the magnetostatic scalar potential to examine a new situation in the end gap.

Technically, let us regard a single unit north pole in the rotor, going from position 1 to position 2 (the acceleration cycle, where the engine will deliver shaft horsepower against a load), and then from position 2 to position 1 (where the magnetostatic scalar potential must be regauged to equal or exceed the potential at position 1, in order for the rotor to continue unabated or even further accelerate. I.e., in the separation gap, a regauging operation must be done so that the "stator to inner" potential is increased equal to or exceeding the "stator to inner" potential of position 1. In normal machines, the energy input part of the cycle is always where the design engineer *forcibly* inputs energy from outside the system to *do physical work* on the machine to *forcibly* "reset" its energy storage back to desired initial conditions. In the past engineers have automatically assumed $COP < 1.0$ without exception, since their *forcible* RESET work was itself not 100% efficient, and so it was always equal to or greater than the maximum theoretical energy output to the load during the motor part of the cycle from point 1 to point 2.

To do better than that, the engine must perform the regauging or RESET of the its energy storage, *without performing tangential "drag" work on the rotor*. For that purpose, an electromagnet is utilized to fill the end gap in the stator, arranged so that when it is activated its north pole (and its B-field) will face radially inward. A small current activates the coil weakly, through a distributor with breaker points. At the proper timing (i.e., when the rotor is directly opposite the electromagnet polepiece, the ignition points are sharply broken in the circuit with the coil of the electromagnet. Momentarily, a very high magnetic scalar potential will appear at the end of the electromagnet's coil as the collapsing field is highly amplified in trying to sustain the previous current in its previous direction. The end result is the nearly instantaneous formation of a strong magnetostatic scalar potential (pole), of north polarity, on

the stator polepiece facing the rotor. Note that no *radial* work can be done on either the stator polepiece or the rotor by this high potential.

The magnetostatic scalar potential in the end gap is now higher than the potential at position one. Consequently a clockwise tangential B-field force exists between the end gap potential and the lower potential at position one. A clockwise tangential force therefore appears upon the rotor, and the rotor is *accelerated* and "boosted" out of the stator gap and back past point 1. Note we are assuming it best to shade the backside of the electromagnet nearest Point 2. This is to reduce a back-drag that would otherwise be created between the increased end gap potential and the potential at Point 2. At that point the electromagnet has lost its potential, but the engine has now been *asymmetrically self-regauged* and now is in the clockwise acceleration field of the rotor-stator permanent magnets.

In short, the rotor perceived the sudden change of magnetostatic scalar potential from the electromagnet in the stator gap as a pseudo-MVP, and the system received a sharp influx of potential energy, with a "free" associated tangential force that can be used to expend the excess potential energy to accelerate the rotor through the regauging zone. To get this, no external work is required to be done on the engine except for that required by the electromagnet's circuitry. Since that energy loss can be made quite nominal by conventional electronic practices, the engine permissibly provides $COP > 1.0$. It can therefore be rigged to power itself and a load simultaneously.

Placed in an electric vehicle with necessary switching circuitry and ancillary equipment, a properly designed Takahashi engine and its derivatives should be capable of starting from a single ordinary battery, then powering the vehicle agilely, powering the accessories, and recharging its own battery — all three simultaneously. It is our understanding that the latest tests of this type engine in an electric motor scooter appears to have demonstrated just these simultaneous capabilities.

There are still some disadvantages with the Takahashi engine. First, the magnets are expensive and would appear to require arduous manufacturing in the asymmetrical configuration shown. The dynamic balancing of the engine would be critical, since the stator is nonsymmetrical and so is the rotor. Also, the single rotor sector would need to be counterbalanced on the shaft, say by another Takahashi stage in counterbalance form. It would be desirable to eliminate these difficulties and also to reduce the rather appreciable costs of such an array of magnets. Nonetheless, the Takahashi engine does appear to be a very viable overunity engine.

Conclusion

In this paper we have briefly discussed the storage and transfer of energy in an electromagnetic circuit from a gauge-theoretic viewpoint. We have presented the multivalued potential and the pseudo-multivalued potential, and their usage in asymmetrically self-regauging the potential in the energy-storing subsystem of an EM engine. Asymmetrical self-regauging (ASR) accomplishes a work-free resetting or "refueling" flow of energy in an electrical circuit, from a modified Poynting vector standpoint, and also provides excess associated forces to utilize that excess energy to do work on the system to increase its kinetic energy. We have discussed embodiments of current-blocking, energy storage, energy shuttling, multivalued potential (MVP), pseudo-MVP, and regauging approach for overunity electrical power systems and for room temperature superconductivity.

In addition we have explained two Japanese overunity engines which have reached or are approaching full production capability in an extremely well-funded, national Japanese strategic effort lasting more than two decades.⁷⁵ We have also discussed the "boosting stator gate" and in the Johnson permanent magnet engine, and its use of a pseudo-MVP to accomplish asymmetrical self-regauging; the gate has been rigorously measured to yield excess acceleration of the rotor approaching it, passing through it, and exiting.

Until recently, in model year 1997 a certain percentage of all new automobiles sold in the U.S. would have been required to be *zero-polluting* vehicles — i.e., *electric* vehicles. That deadline has now been temporarily postponed. U.S. automotive manufacturers are already committed for the specific electric vehicles they will be able to offer in 1997 and later. These U.S. offerings will be bulky, cumbersome, largely impractical, expensive, and maintenance-intensive. They will require frequent and lengthy recharging of their huge battery packs. They will give poor performance, get very low mileage (range) between recharges, and will have only austere power for accessory systems. The manufacturers are likely to have to sell them at a loss in order to meet their mandatory quotas, whenever the deadline is restored.

The Japanese manufacturers appear to be readying for production and gradual introduction of a substantial line of electromagnetic engines which are overunity and self-powered, and a substantial line of powerful electric vehicles utilizing those engines. Those vehicles can be initiated from a single battery and will be self-powered from then on. They will be eminently practical, essentially unlimited in range and performance, can be large and luxurious and agile, can have full-powered accessory systems, and could be expect-

ed to be available in a wide range of sizes and performances.

It appears that Japan has scored a tremendous advance in the field of overunity engines and power systems. Japanese engineers, scientists, and inventors have attacked the problem of their nation's long-standing lack of energy resources with resolve, patience, dedication, and a sustained financial effort. It now appears that their hard work and perseverance has paid off.

For that reason, this paper has been prepared in order to provide the fundamental principles of overunity operation of electrical circuits and rotary electromagnetic engines. By openly publishing this information, which has taken the author decades to produce, it is hoped that the scientific communities of the United States and other nations will be spurred into competitive development. The processes and principles involved do not require exotic materials or processes. No exotic thermodynamics is involved.

With the imminent introduction of practical overunity engines from the Japanese industries and — hopefully — the industries of other nations as well, it appears that the greatest energy, transportation, and industrial revolution in history will now be launched.

FOOTNOTES:

¹Tom Bearden, "A Direct 'Free Energy' Method," 1980. Internal private personal paper at the time, but long since unrestricted.

²T.E. Bearden, "On Rotary Permanent Magnet Motors and 'Free' Energy," *Faust & Zeit*, 1(3), Aug.-Sep. 1989, p. 43-53.

³Johnson's original permanent magnet motor patent is "Permanent Magnet Motor," U.S. Patent No. 4,151,431, Apr. 24, 1979. For a later Johnson patent on his magnetic gating processes, see Howard Johnson, "Magnetic Force Generating Method and Apparatus," U.S. Patent No. 4,877,983, Oct. 31, 1989. In his very complex stator field gates, Johnson created a pseudo-multivalued magnetostatic scalar potential. This allowed the rotor magnet to be given a free "propulsion boost" when passing through the stator gate. *Asymmetrical self-regauging* enabled the stator gate to "attract in" the incoming rotor, then strongly accelerate it on through the gate and out of it during exit. Hard measurements made with an excellent force meter at every 100th of a second verify the multivalued potential boost effect. Johnson is presently processing a separate patent application covering this free "acceleration burst" achieved by means of his Incorporated pseudo-multivalued potential (MVP). It appears that a variety of electrical and magnetic overunity machines actually and unwittingly obtain their overunity operation from (i) the use of an MVP or pseudo-MVP, and (ii) use of one or more of the application rules for utilizing the MVP — without the inventor being aware of his use or partial use of asymmetrical self-regauging.

⁴T.E. Bearden, "The Final Secret of Free Energy," February 1993; released over InterNet and published in several magazines as well.

⁵Thomas E. Bearden and Stephen L. Patrick, "Method and Apparatus for Extracting Internal EM Flow Energy of the Potential and Reducing Back EMF in Electrical Sources to Improve Operational Efficiency," U.S. Patent Application no. 08/192687, Feb. 7, 1994, and Continuation, May 13, 1994.

⁶T.E. Bearden, Steven L. Patrick, and Kenneth D. Moore, "Room Temperature Superconductivity, Poynting Energy Flow, and Overunity Coefficient of Performance of Electrical Systems," U.S. Patent Application, June 7, 1995. Thomas E. Bearden and

Kenneth D. Moore, "Passive and Active Amplification of Input Voltage to Increase Output Field Energy Density of Poynting Generators," Provisional Patent Application, filed July 1995.

⁷T.E. Bearden, "Japanese Overunity Motor and a Commentary," *Explore More!*, Number 13, 1995, p. 37-42. Unfortunately the typesetting process utilized a software translation process from my furnished word processing disk, and it mutilated the mathematical symbology in the published article. The corrected article has now been made available to the magazine's subscribers without charge.

⁸This article was stimulated by a foreign article, "Perpetual Motion Machine," *Guarantist*, Sept. 1994. See also a later article by Tony Edwards, "Accident in lab creates super motor," *The [London] Sunday Times*, Dec. 10, 1995.

⁹Teruo Kawai, "Motive Power Generating Device," U.S. Patent No. 5,436,518; issued July 25, 1995. Filed June 17, 1993. 16 claims and 19 drawing sheets. Patents cited: 24 U.S., 2 foreign.

¹⁰Eugene Mallove, "Claimed Over-Unity Magnetic Motor Awarded U.S. Patent: 318% Mechanical Output/Electric Input," *Infinite Energy*, Sept.-Oct. 1995, p. 40.

¹¹Eugene Mallove, "Japanese Television Program Reveals Industry-Government Program in Energy from Magnetic Materials," *Infinite Energy*, Mar.-Apr. 1995, p. 48-49.

¹²David Scott, "Magnetic 'Wankel' for Electric Cars," *Popular Science*, June 1979, p. 80-81.

¹³We particularly include the nonlinear thermodynamics of a dissipative system far from thermodynamic equilibrium. A very useful summary of the theory of such systems is given by Gregoire Nicolis, "Physics of far-from-equilibrium systems and self-organization," Chapter 11 in *The New Physics*, edited by Paul Davies, Cambridge University Press, New York, 1989, p. 316-347.

¹⁴Indeed, any ordinary power system is just such an open system already, at least while being recharged or refueled, and when actually performing work upon the load or suffering losses. E.g., when we add the fuel to a gasoline engine, we introduce "concentrated stored energy" (the fuel) from an external source. So we periodically "open" the system and inject a form of excess energy — the fuel. While we are adding the fuel, the system is not only open but also out of equilibrium. In other words, rigorously we *asymmetrically regauge* the system by changing its inherent potential energy. Further, the refueling and regauging operation is accomplished by a negligible amount of work with respect to the energy that can then be dissipated as work. Since we already regauge our normal engines, then it behooves us to extract this "master process" and further develop it.

¹⁵The reader is referred to any good texts on particle physics. Two excellent, readable references are by Nobellian T.D. Lee, and they are (i) *Symmetries, Asymmetries, and the World of Particles*, University of Washington Press, Seattle, 1988 and (ii) *Particle Physics and Introduction to Field Theory*, Harwood Academic Publishers GmbH, New York, 1981, Second Printing with Corrections, Nov. 1982. Particularly see p. 181-188 and p. 824-828 of the latter reference.

¹⁶Broken time symmetry and its concomitant violation of local energy conservation was the basis for the overunity mechanism proposed for a Johnson permanent magnet motor in 1989. See Footnote 2 above.

¹⁷As a case in point well-known in solid state theory, adding energy to Dirac electrons can lift "virtual" electrons from the Dirac sea, into real, observable electrons that "appear from nowhere." Further, broken symmetry is involved in the process *a priori*.

¹⁸Fundamentally, broken symmetry involves an asymmetrical alteration (regauging) in the local energy density of the vacuum and its structuring. This corresponds to a locally curved spacetime in some aspects. As a special case, use of an MVP allows the local energy of spacetime to be altered in an instantaneous "jump" between two different uncurved states, but adds a local curvature as well since a force is produced. So one can asymmetrically "regauge" the vacuum's local energy in a chosen "reset" sector of an electrodynamic machine, without having to pay the piper. One can have the machine do it itself. Rigorously it need not require work in order to transfer energy to a system, if the energy transfer is done by lossless nonmaterial energy transport, as in regauging. However, once the

excess energy is received and stored, it requires an associated EM force field to dissipate the stored energy in the form of accelerating work performed upon the rotary EM engine. Consequently, to use the "freely collected energy" to provide COP>1.0, one must either perform the initial regauging asymmetrically, or if it is symmetrical then it must be followed by an asymmetrical regauging.

¹⁹We strongly point out that physicists and engineers do not really know *what electric charge is*. The rare physicist will even openly admit it. E.g., see M.P. Silverman, *And Yet It Moves: Strange Systems and Subtle Questions in Physics*, Cambridge University Press, Cambridge, 1993, p. 127. Quoting: "...curiously enough, we do not know exactly what charge is, only what it does. Or, equally significantly, what it does not do." The present author has proposed the rigorous definition $q = \oint \mathbf{D} \cdot \mathbf{n}$. That definition has profound implications, but these are beyond the scope of the present paper.

²⁰Rigorously, the dipole diverges the vacuum flux into two streams of energy. The conductors (transmission lines) connected to the ends of the dipole provide the gating and transmission medium.

²¹I.e., directly from the charge q 's virtual photon exchange with the vacuum flux.

²²Immediately one notices the temporal difference between what is happening overall in the electron shells of the atoms, as compared to what is happening overall in their nuclei. For one thing, a nucleus is on the average *time-reversed*.

²³To first order an atom can be considered as a dynamic set of such electric dipoles, where each dipole has its negative end as an electron in the electron shells, and its positive end as a positive charge in the nucleus. One thus realizes that the atom is a powerful Poynting generator of very peculiar kind, and intense fluxes of these special and unusual Poynting flows bathe all matter and its vicinity. Further, this "massic energy flux" is internally structured because of the electron shell structuring and dynamics, and the nuclear shell structuring and dynamics, of the "atoms as transmitters and receivers." A novel kind of atomic energy program could be devoted toward directly extracting EM energy from this massic flux by an adapted and extended electromagnetics, without tampering with radioactive residues, huge reactors, and centuries-long pollution and waste disposal problems.

²⁴John D. Krauss, *Electromagnetics*, Fourth Edn., McGraw-Hill, New York, 1992. Pages 577-580 cover the applications for Poynting vector to very simple circuits. Figure 12-60, on p. 578 shows Krauss' diagrams for the Poynting power flow in a simple circuit containing a battery and a resistor. The diagrams err, however, in showing the energy flow originating in the battery and ending in the resistor. In fact the energy flow originates in the vacuum surrounding and permeating the battery, flows out through both terminals and along both feeder lines to the resistor, then is scattered back to the vacuum from the resistor. Some of the flow is also scattered back to the vacuum from the internal resistance of the battery itself. Another very useful reference is H.G. Booker, *Energy in Electromagnetism*, Peter Peregrinus LTD, 1982. Part of the IEE Electromagnetic Waves Series 13.

²⁵This can be seen as follows: $\mathbf{S} = \mathbf{E} \times \mathbf{H}$. There exists an E-field between the separated charges of the dipole. Further, the fundamental charged particles of the charges are spinning, and they have magnetic spins and magnetic fields which are aligned perpendicular to the E-field. Therefore $\mathbf{E} \times \mathbf{H} \neq 0$.

²⁶For mathematical proof that any scalar potential is indeed a hidden bidirectional flow of EM energy, see E.T. Whittaker, "On the Partial Differential Equations of Mathematical Physics," *Mathematische Annalen*, Vol. 57, 1903, p. 333-355. Whittaker mathematically decomposes the scalar potential into a bidirectional series of EM wave pairs in a harmonic sequence. Each wavepair consists of the wave and its phase conjugate.

²⁷It turns out that we already evoke much greater flows of energy in our circuitry than we are able to collect and utilize, the nominal fraction of collection and usage being about 10^{-13} . Could we collect and use more of it, then a single flashlight battery could drive a battleship. Additional discussion of this interesting phenomenon is given in T.E. Bearden, "Chasing the Wild Dragon: Foundations of a New Science," *The Virtual Times*, Internet node WWW.HSV.COM,

²⁸Although Ehrenhaft repeatedly reported such. Recently there has been important new work done in this area that indicates magnetic charge, but is not yet definitive. E.g., see V.F. Mikhailov, "Observation of magnetic monopoles in the field of a line conductor," *J. Phys. A: Math. Gen.*, Vol. 18, 1985, p. L903-L906. See also Mikhailov, *J. Phys. Condensed Matter*, Vol. 5, 1993, p. 351-360.

²⁹Note that when one says "magnetostatic scalar potential," one has also said "magnetic pole" and "pole strength."

³⁰Whenever a potential gradient is applied to a bipolar circuit, a Poynting vector S flow of field energy density then flows along the potentialized conductor and a "reference" level of S flows along the "ground" side conductor. See John D. Krauss, *Electromagnetics*, Fourth Edn., McGraw-Hill, New York, 1992, Figure 12-60, a and b, p. 578 for good diagrams of the Poynting flow in a simple circuit containing a battery and a resistor. As do all electrodynamists, however, Krauss errs in showing the flow as originating in the battery and ending in the resistor. In fact the energy flow originates in the vacuum surrounding and permeating the battery, converges into the battery dipolarity, flows out through both terminals and along both feeder lines to the resistor, then is scattered back to the vacuum from the resistor. Some of the flow is also scattered back to the vacuum from the internal resistance of the battery itself. The Poynting S -flow is outside the conductors, as is well-known, and in the form of field energy flow, via $S = E \times H$, where the conductors act as "guides" or "railroad tracks" for S .

³¹See E.T. Whittaker, "On the Partial Differential Equations of Mathematical Physics," *Mathematische Annalen*, Vol. 57, 1903, p. 333-355. Whittaker mathematically decomposes the scalar potential into a hidden bidirectional series of EM wave pairs in a harmonic sequence. Each wavepair consists of the wave and its phase conjugate. Thus any $\Delta\phi$ (as across the terminals of a dipolar electrical power source) *a priori* involves an ordered, hidden, bidirectional EM wave flow. I.e., $\Delta\phi$ identically is such a hidden bidirectional EM energy flow.

³²It can be shown that normal electrical circuits nominally collect as $J\phi$, and utilize to power the loads and losses, only about 10^{-13} of the actual Poynting energy flow $S = E \times H$ that they evoke.

³³Asymmetrical self-regauging collects just a tiny bit more of the raging S -flow that is flowing outside the conductors. E.g., if the self-regauging EM engine collects as $J\phi$ some 10^{-12} of the S -flow instead of the more usual 10^{-13} , it can have a COP = 10. And so on.

³⁴J.D. Jackson, *Classical Electrodynamics*, 2nd Edn., John Wiley & Sons, New York, 1975, p. 220-223.

³⁵So-called "canceling" oppositive EM fields are actually produced, which sum to a vector zero which the electrodynamists discard by assumption. We point out, but do not further pursue, that the locally-produced field energies of the oppositive fields remain and add, even though the fields offset each other translationally, since the energy of the field is proportional to its square, and that is always positive regardless of field orientation. Thus "trapped" EM energy has been localized in spacetime in the symmetrical regauging of CEM, and this rigorously is a gravitational change and therefore a local curvature of spacetime *a priori*. The electrodynamists in their symmetrical neatness have ignored the fact that gauge symmetry electromagnetic transformations are accompanied by local gravitational changes. Heaviside became aware of this type of relationship, and worked out a theory of electrogravitation based on his S -flow theory (which added closed-loop circulations of energy which vectorially summed to net zero). Those "trapped EM energy flow loops" were gravitational, in Heaviside's theory. His hand-written notes on this theory were found long after his death, beneath the floorboards of his little garret apartment. See H.J. Josephs, "The Heaviside papers found at Paignton in 1957," *The Institution of Electrical Engineers Monograph No. 319*, Jan. 1959, p. 70-76. Nothing further seems to have been done with Heaviside's testable draft theory of electrogravitation.

³⁶Physicists love symmetry, turn to it at every opportunity, and will do almost any mathematical manipulation to obtain it, because they regard it as "beautiful." And so it is. Performing work, however, requires breaking symmetry to produce a force, and

then using that force to do the work. If we wish to build a system that produces more work output than the work that we ourselves have to do upon it to run it, then *a priori* that system must be asymmetrical and — to many physicists — "ugly."

³⁷Many theoreticians object to the inclusion of asymmetrical regauging, preferring to consider any regauging as symmetrical. As we showed, the familiar S -regauging is actually comprised of two separate but simultaneous asymmetrical regaugings (A-regaugings). All we need do is (1) break the simultaneity, or (2) break the "canceling" capability of the two A-regaugings, or (3) both.

³⁸Again we strongly stress that asymmetrically regauging a potential in the system also regauges the local vacuum potential and the intensity of the vacuum potential's constituent bidirectional EM energy exchange with the system. It directly alters the stored energy of the system. See again Whittaker, 1903, *ibid*. to see this clearly.

³⁹E.g., see G.A. Kotelnikov, "Toward a nonlinear electrodynamics," *Izvestiya Vuz Fizika*, 38(2), Feb. 1995. The English translation is in *Russian Physics Journal*, 38(2), Aug. 1995, p. 205-208. We point out that exploration of this "infinite set" of Maxwell equations is still in its infancy. Nonetheless its conclusions already can be quite startling to the classically trained engineer.

⁴⁰To see how really complex are the actions in motors and generators, the reader is referred to the work of Gabriel Kron, possibly the greatest U.S. nonlinear electrical scientist of all time. Even full general relativity — which Kron rigorously applied to rotating electrical machines — still falls short of what is needed. E.g., the technical reader might wish to peruse his "Four abstract reference frames of an electric network," *IEEE Transactions on Power Apparatus and Systems*, PAS-87(3), Mar. 1968, p. 815-823. Electrical engineers often treat their stationary networks, rotating machines, and microwave electronic devices as a collection of impedance elements Z without decomposing Z into its RLC components. Kron shows that a lumped or distributed impedance network, surrounded by its own electromagnetic field, is actually the sum of four different types of multidimensional networks: (1) the well-studied 1-network of branches in which the currents flow, (2) a 0-network formed by all the point generators, (3) a 2-network of equipotential surfaces that pass through the generators perpendicularly to the branches, and (4) a 3-network composed of three-dimensional impedance blocks surrounding the branches. Thus the topological structure of a stationary or rotating, electric or electronic network is neither a graph nor a polyhedron, but a so-called fiber bundle over a non-Fleemannian manifold.

⁴¹For one thing, the electromagnetics operations one can accomplish are to a large extent dependent upon the topology of the mathematics in which one's electromagnetics model is embedded. Normal EM theory utilizes vector or tensor algebra, but many higher topology algebras are in fact available. E.g., normal analysis by vectors and tensors cannot show many of the operations that Tesla actually accomplished in his patented circuits. Yet an analysis of the circuits in quaternion algebra — which is of higher topology than either vector or tensor algebras — will show these functions clearly. For rigorous proof, see T.W. Barrett, "Tesla's Nonlinear Oscillator-Shuttle-Circuit (OSC) Theory," *Annales de la Fondation Louis de Broglie*, 16(1), 1991, p. 23-41. We point out that Barrett is a fine theoretician and one of the pioneers of ultrawideband radar — which does not function by the ordinary "sine wave decomposition" and transform methods of the conventional EM radar theory.

⁴²In a general sense, modern gauge theories are an extensive expansion of their parent electromagnetic theory. A particularly good little summary of gauge theories is provided by Benjamin W. Lee, "Gauge Theories," *Encyclopedia of Physics*, eds. Rita G. Lerner and George L. Trigg, Addison Wesley Publishing Company, Reading, MA, 1981, p. 349-351. A useful and slightly more technical summary is given John Taylor, "Gauge theories in particle physics," Chapter 17 of *The New Physics*, edited by Paul Davies, Cambridge University Press, New York, 1989, p. 458-480.

⁴³For just a "quick feel," see "Gauge Theory," *Encyclopaedia Britannica*, 15th edition, 1985, Micropaedia vol. 5, p. 147. For a better look, see par-

ticularly, Article 6.5, "Gauge Transformations, Lorentz Gauge, Coulomb Gauge," in J.D. Jackson, *Classical Electromagnetics*, 2nd Edition, John Wiley & Sons, 1975, p. 220-223.

⁴⁴E.T. Whittaker, "On an Expression of the Electromagnetic Field Due to Electrons by Means of Two Scalar Potential Functions," *Proceedings of the London Mathematical Society*, Series 2, Vol. 1, 1904, p. 367-372.

⁴⁵In 1903 Whittaker also showed that any scalar potential is a special set of multiple EM wavepairs. It follows that the interference of two scalar potentials is just the hidden interference of their biwave internal components. In that manner, scalar potential interference is not an oxymoron but just a special case of multiwave interferometry, one in which the interfering waves are "infolded" in 3-space and nonobservable, but their interference becomes observable in ordinary 3-space. The practical ramifications of such scalar interferometry are enormous but beyond the scope of this paper. They have been extensively covered by the author elsewhere.

⁴⁶By "sufficient abruptness" we mean that one fundamental "charge" element or "charge grain" — or in the case of a magnet, one single magnetic domain — has a greater diameter than the distance in which the abrupt change or "jump" of potential occurs. For electrical charges this is difficult, but not for magnets. The "back drag" magnetic work experienced by this "real physical charge" of finite magnitude in a magnetic stator/rotor arrangement varies from the theoretical "point-unit north pole value" used in conventional field theory, as rigorously proved by Howard Johnson in actual force and time measurements. One will just get a small back impulse $\int F(t)dt$, over a very short time dt , which means that less retardation work is experienced in the "jump" or regauging region than the work that was obtained from the unit during the rest of the cycle involving the boost.

⁴⁷Note, however, that by regauging we have great latitude as to how we go about inputting the energy utilized by the system in its operation. We are *not* required to use "fields only" and we are *not* required to do work to input the energy!

⁴⁸Because of the multiple bodies involved, a planetary orbit is actually chaotic rather than exactly linearly oscillatory as shown. But the ideal orbit is linearly oscillatory.

⁴⁹Note that energy is conserved from the sourcing circuit to the load circuit, but work (energy dissipation) need not be conserved, because we are passing energy from a nearly-free source as pure energy transport, bridged into a separate dq/dt current loop containing the load. Almost all of the work (i.e., dissipation of the energy flow) occurs in the load circuit, not the sourcing circuit.

⁵⁰We point out that voltage is joules per (collecting) coulomb. That is, one volt of potential established from one coulomb of charge q to another reference potential level will result in the collection of one joule of excess potential energy upon that coulomb of charge. When we "potentialize" the conducting electrons in a dq/dt -blocked circuit, we freely add real energy to those electrons. Each one corresponds to having been raised from D to A automatically, in Figure 3, as shown by the jump in Figure 4.

⁵¹This description of course is for the ideal system. In a real system, a very small current must be drawn in the primary source loop. However, this current may be made much smaller than the load current. Also, voltage multipliers can be added in the receiving load loop to increase the load voltage and current, without affecting the primary source current and dissipation.

⁵²See any good EM text, in its section on Poynting vector theory. For the original, see J.H. Poynting, "On the transfer of energy in the electromagnetic field," *Philosophical Transactions of the Royal Society of London*, Vol. 175, 1884, p. 343.

⁵³For a statement of Heaviside's precedence, see Editorial, "The Transfer of Energy," *The Electrician*, Vol. 27, July 10, 1891, p. 270-272. Quoting: "...the idea that energy is located at all, and that, when it changes its position, it must move along a definite path, is quite a new one. The law of the conservation of energy implies that energy cannot disappear from one place without appearing in equal quantity somewhere else; but, although this fact has long been accepted, it is only within the last few years that the

idea of transference of energy has been developed, or that anyone has attempted to trace out the actual path along which energy flows when it moves from place to place. The idea of an energy current is of more recent date than the electro-magnetic theory, and is not to be found explicitly stated anywhere in Maxwell's work. We believe that the first time it was applied to electrical theory was in the pages of *The Electrician*, by Mr. Oliver Heaviside, to whom so much of the extension of Maxwell's theory is due. The idea was also independently developed and brought to the notice of the Royal Society in a Paper by Prof. Poynting." As quoted in Nahin, Paul J. (1988) *Oliver Heaviside: Sage in Solitude*, p. 117.

⁵⁴Heaviside's theory was actually superior to that of Poynting, because Heaviside included a term which later allowed him to create a theory of electrogravitation. For an article showing what Heaviside's unpublished papers accomplished in this respect, see E.R. Laithwaite, iMiles Walker — a Pioneer at Met-Vick and UMIST *Electrical Review*, 211(16), Nov. 12, 1982, p. 44-45. To Poynting's power equation $P = E \times H$, Heaviside added an arbitrary vector G , representing a circual energy flux, to obtain $P = E \times H + G$. Note that Heaviside is correct. In transferring excess Poynting field energy density flow to closed current loops and coupling the field energy density to the conduction charges in that closed current loop, our patent pending Poynting generator circuits provide additional trapped EM energy that is collected in the receiving current loop prior to its dissipation in the load. It follows that these patent-pending circuits can be *electrogravitational* circuits if so utilized with significant power and in the proper frequency regime (gravitation is very much enhanced at ELF frequencies). As an example, the force existing between two one-coulomb charges separated by one meter distance is about one million tons. So if one is significantly gating the Poynting energy flow in a deterministic manner, one can engineer and gate electrogravitational forces. Obviously the manipulation of local gravity and antigravity becomes a factor in overunity electrical systems when even fractions of such powerful forces are generated. Therefore it should not be surprising that, in addition to overunity operation, the Sweet vacuum triode amplifier operating at 60 Hz was reported to exhibit strong antigravity effects when specially rigged to do so. See Floyd Sweet and T. E. Bearden, "Utilizing Scalar Electromagnetics to Tap Vacuum Energy," *Proceedings of the 26th Intersociety Energy Conversion Engineering Conference (IECEC '91)*, Boston, Massachusetts, 1991, p. 370-375. According to data published by the authors, the machine incrementally lost 90% of its weight when its power output was incrementally raised to 1 kilowatt.

⁵⁵See J. Slepian, "Energy and energy flow in the electromagnetic field," *Journal of Applied Physics*, Vol. 13, 1942, p. 512-518; — "Energy flow in electric systems — the VI energy flow postulate," *Trans. Amer. Inst. Elect. Engrs.*, Vol. 61, 1942, p. 835-841. Slepian proposed some eight vectors which satisfied Poynting's theorem. For a critique of alternative Poynting vectors, see P.C. Peters, "Objections to an alternate energy flow vector," *American Journal of Physics*, 50(12), Dec. 1982, p. 1165-1168.

⁵⁶As an example, see C.J. Carpenter, C.J., iElectromagnetic energy and power in terms of charges and potentials instead of fields, i *IEEE Proceedings A (Physical Science, Measurement and Instrumentation, Management and Education)*, (UK), 136A(2), Mar. 1989, p. 55-65. Carpenter advocates replacing the Poynting vector S with the Slepian vector $j\phi$. He does not grasp the difference between the field energy flow evoked by our circuits along the conductors and exterior to them, and the incredibly tiny fraction of that energy that is actually collected as $j\phi$. He is correct, of course, in stating that $j\phi$ prescribes the energy transport that is actually utilized (dissipated) by the circuit.

⁵⁷As an example of one of the very best of the theorists who does struggle with Poynting flow in circuits, see John D. Kraus, *Electromagnetics*, 4th Edn., McGraw-Hill, 1992, p. 575-580.

⁵⁸See any good electromagnetics textbook or handbook. E.g., see E.U. Condon, i4. Poynting Theorem, iChapter 1: Electromagnetic Waves, i *Handbook of Physics*, 2nd Edn., Eds. E.U. Condon and Hugh Odishaw, McGraw-Hill, New York, 1967, p. 6-6 to 6-7.

⁵⁹We accent again that a "flow of potential" or "flow of voltage" is automatically a flow of S .

⁶⁰E.g., see Mark A. Heald, iElectric fields and charges in elementary circuits, i *American Journal of Physics*, 52(6), June 1984, p. 522-526. Quoting: "The charges on the surface of the wire provide two types of electric field. The charges provide the field inside the wire that drives the conduction current according to Ohm's law. Simultaneously the charges provide a field outside the wire that creates a Poynting flux. By means of this latter field, the charges enable the wire to be a guide (in the sense of a railroad track) for electromagnetic energy flowing in the space around the wire. Intuitively one might prefer the notion that electromagnetic energy is transported by the current, inside the wires. It takes some effort to convince oneself (and one's students) that this is not the case and that in fact the energy flows in the space outside the wire."

⁶¹Stoney, G. J. (1897) "On a Supposed Proof of a Theorem in Wave-motion, Letter To the Editors, *Philosophical Magazine*, 5(43), 1897, p. 368-373.

⁶²Whittaker, 1903, *ibid*.

⁶³Arbitrary elimination of this interior ordering would seem to be where present quantum mechanics seriously errs. Because the scalar potential can be mathematically composed into hidden dynamic order and organization in the wave mathematics, then the particle mathematics of the interior must also show hidden order. Instead, quantum field theory essentially regards the virtual particles constituting the interior flux as randomly varying, because of the application of Gibbs' thermodynamic statistics. By removing the hidden order, quantum mechanics "loses" the macroscopic world, because integrated randomness is still randomness. This is the reason for the critical foundations problem in QM today: that of the missing chaos (the missing order). Actually the statistics should be changed to chaotic statistics, in which case hidden order is automatically incorporated and the chaos is no longer missing. However, that petrifies most quantum physicists, because it shows that quantum change itself — and therefore physical reality — can be directly engineered, even before the collapse of the wave function and the emergence of the physical change as an observable. Stoney and Whittaker's work is key to that engineering of the "hidden inner EM wave mechanics." I.e., to *engineerable* hidden variable theory.

⁶⁴To drive a primary of a transformer in this fashion, the circuit feeding the excess voltage and emf to the primary must be isolated from the reflected impedance of the transformer. A complete dq/dt blocker, such as the Fogal semiconductor, in the conducting lines accomplishes this insulation from the reflected impedance. In short, one must transfer voltage and emf between two closed current loops, absolutely without the exchange of dq/dt between the loops, and without the reflectance of impedance between them.

⁶⁵The practical difficulty for circuits, however, is that the usual inductive and capacitive time delays vanish or seriously decrease. The formation, collapse, or escape of such a "magnetic field that is unattached to i " occurs at light speed as it does in the vacuum, via free (uncollected) field energy S -flow rather than sluggish $J\phi$ collected energy flow speeds, whenever the $J\phi$ is blocked. One must therefore utilize microwave switching techniques rather than the far slower switching techniques such as usually utilized in motors and generators, which will gain nothing. The rule is this: Coupling of the potential fields to sluggishly moving electrons accounts for the gradual buildup and collapse (i.e., the hysteresis) of inductors. In other words, it is the involvement of $J\phi$. When the electron movement as dq/dt is blocked, the field energy moves as radiant energy and S -flow, hence essentially at the speed of light. There is no $J\phi$, and hence no hysteresis delay. As may be seen, the use of coils, transformers, and capacitors under dq/dt -blocking conditions is an art unto itself.

⁶⁶Another difficulty in working with dq/dt -blocked circuits is that, in the interior Whittaker structuring of the scalar potential, nonlinear phase conjugate optics effects apply, regardless of frequency. Since potentials extend outward from their source, in a circuit the semiconductors — which are all made of optically active nonlinear materials — thus have a hidden interior Whittaker channel inside the potentials connecting them. Self-targeting (iterative amplified retroreflection) in the interior channel between these semiconductor

components easily leads to serious, even explosive, damage to them. Again, this technology is an art unto itself and still largely undeveloped. My colleagues and I are in process of filing another patent in this area, where the difficulties are partially controlled.

⁶⁷Obviously here we are considering an *ideal* transformer. A real transformer will inevitably experience some losses in the transformer itself.

⁶⁸We strongly emphasize that all closed current loops furnish their own conduction charges for their current dq/dt . The source coupled into such a circuit does not furnish a single electron, but only excess voltage and emf. For a simple battery/resistor circuit, e.g., for every electron that is emitted from the positive terminal of the battery into the external circuit, a corresponding (but energetically spent) electron enters the negative terminal. No current dq/dt is actually "drawn from the battery" per se! Instead, the free conduction electrons already in the circuit are *potentiated* by the battery's emf, which drives the electrons through the circuit, including through the loads. The dq/dt acts rather like incompressible fluid flow; for one electron to fall through the entire potential drop in the external circuit from the positive line to the ground return line, a "spent" electron in the ground return line must be forcibly pushed back up through the battery itself, against the battery's back emf. If one shuttles-in and couples excess voltage and emf to a closed current loop, *regardless of how it is accomplished*, the loop is potentiated and regauged. That loop will thus be "powered" without furnishing current dq/dt to it, and without dissipating the primary source furnishing the excess field energy density (voltage) and emf.

⁶⁹As is well-known in quantum physics, the vacuum is already filled with seething EM energy, and is thus a legitimate energy source. Further, the conventional protest that thermodynamics prohibits extracting and using the vacuum's electrical energy as heat and power is unfounded. Thermodynamics in fact permits just such extraction and usage. For proof, see Daniel C. Cole and Harold E. Puthoff, "Extracting Energy and Heat from the Vacuum," *Physical Review E*, 48(2), Aug. 1993, p. 1562-1565. So one no longer has to prove that the source is there, or that it is *possible* to extract and use the energy. One just has to show how the extraction, gating, collecting, and usage of the vacuum energy is accomplished.

⁷⁰We may also regard this exchange of energy between the vacuum and the bipolarity (potential) of the source to be a wave exchange, as shown by Whittaker, 1903, *ibid*. Any scalar EM potential can be mathematically decomposed into a harmonic series of EM wavepairs, where each pair consists of an EM wave and its true phase conjugate (time-reversed) replica wave.

⁷¹E.g., as in a charge-saturated nonlinear ferroelectric capacitor of the 1950s kind. The voltage on such a capacitor can be varied without varying the charge, and the collected energy varies as Vq . In this case there is no dD/dt and there is no dq/dt , but there is dq/dt .

⁷²Note that an exciting feature of this equation is that, when applied to a static Poynting generator (i.e., crossed static E -field and H -field, then together with Poynting's vector definition it proves that the static generator produces an unending stream of free energy flow from the vacuum. So any static Poynting generator is also a free energy source. Both a static electric dipole and a static magnetic dipole are such static Poynting generators and therefore free energy sources.

⁷³Teruo Kawai, "Motive Power Generating Device," U.S. Patent No. 5,436,518, July 25, 1995. Filed June 17, 1993. 16 claims, 19 drawing sheets.

⁷⁴Note that the patent contains an error, stating at first that the torque and rotation are counterclockwise, then showing it moving correctly clockwise from figure to figure in the series.

⁷⁵In comparison, the Kawai engine is much more symmetrical and straightforward than the Takahashi engine. No costly array of stator magnets is required. The stator need not be a costly spiral buildup. Maintenance, reliability, and reduced cost of production should be much improved as compared to the Takahashi engine.